The semantics and pragmatics of multi-head comparatives

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Abstract

A **multi-head comparative** (MHC) contains multiple comparative expressions: e.g., *Fewer people own more wealth in this country than anywhere else* (Chomsky 1981/1993, von Stechow 1984).

This paper argues that such an MHC is closely related to **cumulative-reading sentences** like *At most 3% of the population own at least 70% of the wealth* (see Krifka 1999, Brasoveanu 2013): the former is the comparative form of the latter, expressing multi-dimensional comparisons based on multi-dimensional measurements.

I further argue that the multiple dimensions involved are not interpreted independently. Rather, they jointly define a **composite dimension** that addresses a **single underlying degree Question under Discussion** (**QUD**). In the examples above, the relevant QUD concerns a composite degree of unevenness, which is reflected by the **contrast** between a **small** share of the population and a **large** share of the wealth, or by the **contrast** between a **decrease** along the population dimension and an **increase** along the wealth dimension.

This composite degree of unevenness plays a crucial role in determining what is measured or compared: not the entire population and all the wealth, but the population and wealth shares that together most strikingly (i.e., informatively) represent the degree of unevenness. Thus the study of MHCs brings new insights into comparatives and the role of QUDs and informativeness in natural language.

Keywords: Comparatives, Multi-head comparatives (MHCs), Multi-measurement sentences, Cumulative-reading sentences, Degree semantics, Degree questions, QUD, Scales, Dynamic semantics, Maximality, Informativeness

1 Introduction

A **multi-head comparative** (MHC) contains multiple comparative expressions. (1)–(3), discussed by von Stechow (1984), are representative examples from the literature. (4)–(6) are naturally occurring examples drawn from *the Corpus of Contemporary American English* (https://www.english-corpora.org/coca/, Davies 2008-).

(1) More silly lectures have been given by more boring professors – than I would have expected. (von Stechow 1984: p.42, (136a); Chomsky 1981/1993: p.81, §2.4.4, (6ii))

(2) Less land produces more corn than ever before. (von Stechow 1984: p.46, (156))

(3) No airline saves you more money in more ways than Delta. (Ibid., p.46, (157))

(4) Henry Ford's assembly line in 1913 increased productivity. Fewer people made more stuff. (2002, NEWS: *Atlanta Journal Constitution*)

(5) Fewer and fewer people own more and more of the country's land. (2001, MAG: *Christian Century*)

(6) As I moved through the years, I saw that I would make more money in more skilled positions. (2012, BLOG: pjmedia.com)

Although MHCs have been noted for decades in formal syntax and semantics (see, e.g., Chomsky 1981/1993, von Stechow 1984), a thorough investigation is still lacking. On the one hand, von Stechow (1984) suggests that their semantics should follow naturally from existing theories of comparatives:

> 'I don't think that these cases represent genuine semantic problems. They can essentially be treated with the methods we already have at our disposal. But it took me quite a while to realize this, because the examples are conceptually rather complicated and are neglected in the literature.'
>
> (von Stechow 1984: pp.41–42)

On the other hand, the grammaticality and interpretability of MHCs have been much debated (see, e.g., Hendriks 1994, Hendriks and De Hoop 2001). Even von Stechow, while claiming that MHCs do not pose genuine semantic problems, acknowledges that such examples are 'conceptually rather complicated.'

Indeed, not all sentences containing multiple comparative expressions are interpreted in the same way. The examples in (1)–(6) are intuitively natural and readily understood. They combine changes along two dimensions to yield a coherent assessment of a contextually salient complex issue, e.g., teaching quality, corn productivity, airline value for money, assembly-line efficiency, economic inequality, or career success.

By contrast, (7) constitutes a controversial case. Although (7) is discussed alongside the MHC examples in (1)–(3) by von Stechow (1984), many native speakers find it difficult to interpret and regard it as degraded. In addition, a sentence like (8) has multiple comparative expressions and is acceptable, but it does not seem to address any complex issue in the way that (1)–(6) do.

(7) %<u>More</u> dogs ate <u>more</u> rats than cats ate mice. (von Stechow 1984: p.43, (140))

(8) <u>Nine more</u> schools choose a <u>longer</u> school year than before. (They join 20 elementary schools on an extended calendar.) (adapted from https://www.aps.edu/news/archives/news-from-2021-2022/9-more-aps-schools-choose-a-longer-school-year)

What distinguishes MHCs in (1)–(6) from cases like (7) and (8)? How are MHCs in (1)–(6) interpreted? This paper aims to answer these questions.

Inspired by Krifka (1999), I draw a parallel between **cumulative-reading** sentences like (9) and MHCs. Cumulative-reading sentences express **measurements** along multiple dimensions, while MHCs express **changes** along multiple dimensions. Thus an MHC is the comparative form of its corresponding multi-measurement sentence.

(9) In Guatemala, at most 3% of the population own at least 70% of the land.

I argue that in both constructions, the multiple dimensions involved are not interpreted independently. Rather, they jointly define a **composite dimension** that addresses a **single underlying degree Question under Discussion** (QUD).

For example, sentence (9) concerns not the measurements of the entire land-owning population or their total landholdings, but the degree of **unevenness** – reflected by a small share of the population and a large share of land owned by them. Likewise, the MHCs in (1)–(6) concern not two independent changes, but a change along a composite scale, characterized by the **contrast** between a **decrease** and an **increase** (e.g., *less . . . more*), or by **parallel changes in the same direction** (e.g., *more . . . more*).

In brief, what distinguishes MHCs like (1)–(6) from other cases is the availability of a contextually salient **degree Question under Discussion** (QUD, Roberts 1996/2012), whose resolution involves integrating multiple dimensions into a composite scale (e.g., low price and discount flexibility jointly determine value for money in (3)).

In the interpretation of MHCs, the relevant degree QUD and its associated composite scale play a crucial role in determining what undergoes comparison: not necessarily the maximal value along each dimension, but rather what is associated with QUD-based maximal informativeness along the composite scale (e.g., the population and wealth shares that together reflect the most striking degree of social inequality in a country). Although this QUD and its composite scale are often not uniquely recoverable from an MHC, the polarity of comparative expressions (e.g., *less*, *more*) provides clues to the monotonicity of the composite scale and thereby restricts compatible QUDs.

The proposed analysis of MHCs integrates existing insights on cumulative-reading sentences and comparatives, largely supporting von Stechow (1984)'s view that MHCs do not pose substantial challenges to semantic theory. At the same time, it sheds new light on the role of degree QUDs and QUD-based informativeness in the interpretation of multi-dimensional measurements and comparisons, demonstrating how sentence interpretation is guided by a conversational goal.[1]

Below, §2 reviews the empirical observations based on Chomsky (1981/1993), von Stechow (1984), Hendriks (1994), and Meier (2001). §3 connects MHCs with cumulative-reading sentences, drawing on insights from Krifka (1999) and Brasoveanu (2013). §4 develops a dynamic semantic analysis that compositionally derives the meaning of MHCs. §5 addresses theoretical implications. §6 concludes.

2 Empirical observations

I start with a typical MHC example and von Stechow (1984)'s analysis (§2.1). §2.2 presents MHC variants. §2.3 reviews degraded cases and the accounts proposed to explain them, with §2.3.1 discussing the syntax of MHCs. This section establishes the empirical scope of the paper and sets the stage for the proposed analysis. The connection between MHCs and cumulative-reading sentences will be developed in §3.

[1] Nouwen (2024)'s work on meiosis and hyperbole as well as Greenberg (2018) and Zhang (2022)'s works on *even* also draw on hidden, contextually salient scales that serve conversational goals. See §5.1.

2.1 A typical MHC and its interpretation

According to von Stechow (1984) (p.45, (153)), a typical MHC like (10) expresses **two independent comparisons**: one comparing the number of silly lectures (see (10a)), and the other comparing the number of boring professors (see (10b)).

(10) <u>More</u> silly lectures have been given by <u>more</u> boring professors – than I would have expected. (=(1), von Stechow 1984: p.42, (136a))

a. **Comparison 1**:
the actual number of silly lectures <u>given by boring professors</u>
> my expected number of silly lectures <u>given by boring professors</u>

b. **Comparison 2**:
the actual number of boring professors <u>that have given silly lectures</u>
> my expected number of boring professors <u>that have given silly lectures</u>

von Stechow (1984)'s analysis also suggests a **mutual restriction** between silly lectures and boring professors, both in the matrix clause and in the *than*-clause (see the underlined restrictions in (10a) and (10b)). Thus, in interpreting this MHC, it is not the total number of silly lectures or the total number of boring professors that is compared with what I would have expected. Silly lectures given by graduate students practicing teaching, as well as boring professors who did not give any lectures, are excluded from the comparison. As will be discussed in §3.1, this mutual restriction between the two noun phrases in (10) is also the hallmark of cumulative-reading sentences.

von Stechow (1984) further points out that the MHC in (10) does not have the event-counting reading shown in (11), i.e., a comparison between the total number of silly-lecture–boring-professor pairs in reality and my expected number.

(11) The number of pairs $\langle x, y \rangle$ in reality > my expected number of pairs $\langle x', y' \rangle$ (where $\textsc{lecture}(x), \textsc{prof}(y), \textsc{lecture}(x'), \textsc{prof}(y'), \textsc{give}(x, y), \textsc{give}(x', y')$)

This event-counting interpretation is too weak. Under the scenario in (12), where the MHC in (10) is intuitively judged false, the reading described in (11) nevertheless comes out as true, contrary to intuition.

(12) Scenario: In reality, 2 boring professors each gave 5 silly lectures; what I expected is that 3 boring professors each gave 3 silly lectures. (Thus $10 > 9$).

2.2 MHC variants

First, the typical MHC pattern illustrated in (10) contains two increases, but MHCs may also involve **changes in opposite directions** that together express a contrast:

(13) Less land produces more corn than ever before. (= (2); see also (4) and (5))

Second, MHCs are **not limited to cardinality comparisons**, but can involve comparisons of **percentages**. In naturally occurring examples (14) and (15), expressions like *trend*, *market share*, and *than anywhere else in the world* strongly favor the interpretation that compares percentages rather than cardinalities:

(14) The trend is indisputable: Fewer people own more of the overall wealth, and fewer companies own more market share.
(https://www.deseret.com/opinion/2020/9/14/21436415/guest-opinion-america-capitalism-strengths-dark-side-too-far-inequality-divisiveness-wealth-gap)

(15) Fewer people own more of the land in Brazil than anywhere else in the world.
(https://glenmorangie.newint.org/features/2003/01/05/cutting)

MHCs like (16) are similar to regular comparatives like (17) in allowing both a cardinality reading and a percentage reading.

(16) Fewer and fewer people own more and more of the country's land. (=(5))
cardinalities ✓; percentages ✓

(17) More people are hired here than elsewhere. **cardinalities ✓; percentages ✓**

Third, comparative expressions in MHCs are **not limited to plural or mass nouns** (cf. von Stechow 1984's restriction). As illustrated by *smaller* in (18), MHCs can compare along dimensions other than quantity. This sentence addresses a technological advance that overcomes the trade-off between quantity and space.

(18) Newer generations of microchips contain more electronic switches on a smaller surface. (Hendriks 1994: (14))

Fourth, like regular comparatives (e.g., (19)), MHCs do **not always contain an overt *than*-constituent** (e.g., (18)). In the absence of an overt *than*-constituent, Lucy's height in

(19) and the number of electronic switches and the size of surfaces in (18) are compared with contextually supplied comparison standards (e.g., Mary's height in (19)).[2]

(19) (Mary is less than 6 feet tall. / Mary is tall.) Lucy is taller. **Regular comparative**

To sum up, MHCs exhibit the same range of variation as regular comparatives: (i) the changes involved may be increases or decreases; (ii) comparisons can be based on cardinalities or percentages; (iii) the compared dimensions are not limited to quantities; and (iv) comparison standards may be overtly expressed by a *than*-constituent or supplied by the context. These observations suggest that the source of degradedness in unacceptable cases lies elsewhere.

2.3 Insights from degraded examples

Various degraded examples have been discussed in the literature, and a range of syntactic, semantic, and pragmatic accounts have been proposed. I argue that, compared with regular comparatives, MHCs pose no additional syntactic or semantic challenges to existing theories of comparatives. Rather, what distinguishes genuine, interpretable MHCs from other cases is pragmatic.

2.3.1 Syntactic structure

A detailed and thorough investigation of the syntax of comparatives and MHCs is beyond the scope of the current paper. Here, based on the syntactic parallelism between regular comparatives and MHCs, I argue that MHCs involve no special machinery beyond what is already needed for regular comparatives.

For comparatives with a *than*-clause (i.e., a clausal comparative), the canonical syntactic analysis assumes ellipsis and a derivation reminiscent of *wh*-movement for the *than*-clause (see (20); see Bresnan 1973, 1975, Chomsky 1977).[3] The assumption of an

[2]Comparatives without an overt *than*-constituent are known as *incomplete comparatives* (Schwarzschild 2010, Li 2023), *contextual comparatives* (Beck et al. 2004, 2012, Hohaus 2015), or *discourse comparatives* (Hendriks 1994). They are distinct from *implicit comparatives*, which use the positive form of a gradable adjective instead of the comparative form (Kennedy 2007, Sawada 2009): e.g., *Compared to Mary, Lucy is tall*.

Incomplete comparatives shed light on the pragmatics and anaphoricity in interpreting comparatives: e.g., in (19), Lucy's height is compared with Mary's – a discourse-salient alternative, not with the measurement of 6 feet or the threshold for *tall* (see Li 2023, Zhang and Zhang-Yukun 2025). See also §2.3.3.

[3]I set aside phrasal comparatives (e.g., *Lucy is taller than me*), which are distinct from clausal comparatives (Hankamer 1973, Heim 1985, Larson 1988) and not relevant to the current discussion.

elided gradable adjective is motivated by evidence from subcomparatives (see (21)). With lambda abstraction at LF, the *than*-clauses in (20) and (21) correspond naturally to degree questions like *how tall the door is* and *how tall Mary is*.[4] At the matrix level, comparative expressions denote a comparison: an increase (with *more*/*-er*) or a decrease (with *less*) based on the degree values that resolve these degree questions.[5]

(20) Lucy is taller [than Mary is ___ ~~tall~~]. **Assuming an elided gradable adjective**
LF of the *than*-clause: [than λd. Mary is d-tall] $\rightsquigarrow$ *how tall is Mary*

(21) The bathtub is wider [than the door is ___ tall]. **Subcomparative**
LF of the *than*-clause: [than λd. the door is d-tall] $\rightsquigarrow$ *how tall is the door*

Interest in MHCs in the literature can be traced to Chomsky (1981/1993), who draws attention to the contrast between (22a) and (22b). He also notes that this contrast is not specific to MHCs but arises more generally, as illustrated by the contrast in (23).

(22) a. [More silly lectures] have been given by [more boring professors] – than I would have expected.
b. *[More silly lectures] have been given by [more boring professors] – than I met yesterday.

(23) a. Pictures of [more people] are for sale – than I expected.
b. *Pictures of [more people] are for sale – than I met yesterday.

According to Chomsky (1981/1993), the unacceptability of (22b) and (23b) results from a violation of the subjacency constraint on *wh*-movement. By contrast, acceptable comparatives (22a) and (23a) involve an anaphoric dependency between the *than*-clause and bracketed comparative expressions, which is not subject to subjacency. Why this anaphoric dependency is unavailable for (22b) and (23b), however, is not explained.

In line with Chomsky (1981/1993)'s ideas, von Stechow (1984) proposes reconstructing covert *than*-constituents so that each comparative expression in an MHC is associated with its own *than*-constituent. For a sentence like (24), von Stechow (1984) claims that the reconstruction gives rise to two possible interpretations of the *than*-phrase (see (24a) and (24b)). Since (24) is degraded, he concludes that 'our grammar doesn't

[4] See Fleisher (2018, 2020) on the semantic parallel between *than*-clauses and degree questions..
[5] The details of the comparison and compositional mechanism will be discussed in §4.2. For now, I adopt a theory-neutral characterization of comparison and focus on the *than*-clause.

work that way. It seems, then, that the restrictions for the reconstruction of a *than*-phrase are rather syntactic than semantic. (p. 47)'

(24) *A greater man would be a better man than Otto. (von Stechow 1984: p.46, (158))
$\rightsquigarrow$ A greater man (**than Otto**) would be a better man **than Otto**.
  a. than Otto is a great man (Ibid., p. 47, (162a))
  b. than Otto is a good man (Ibid., p. 47, (162b))

von Stechow (1984)'s account for (24) is challenged by examples like (25), which suggest that a single *than*-clause may contain multiple elided gradable adjectives, supporting an anaphoric connection between multiple comparative expressions and the same *than*-clause.[6] Thus, a reconstruction of covert *than*-clauses like (24) is unnecessary.

(25) He is a greater and better man than I am. $\rightsquigarrow$ than I am ~~a d-great and d'-good man~~

Moreover, Chomsky (1981/1993)'s contrast in (22) further shows that the reconstruction analysis is not only unnecessary, but also incorrect. Under von Stechow (1984)'s reconstruction analysis (p. 46, (154), (155)), acceptable MHC (22a) is analyzed as (26). The degradedness of (22b) would then follow from the ungrammaticality of (27), where *meet* is compatible with nouns like *professors*, but not *lectures*.

(26) More silly lectures **than** ~~λn.~~ **~~I expected that~~** ~~n~~ **~~silly lectures would be given by boring professors~~** have been given by more boring professors **than** λm. **I expected** **~~that silly lectures would be given by~~** m **~~boring professors~~**. (22a)

(27) *More silly lectures **than** ~~λn.~~ **~~I met~~** ~~n~~ **~~lectures yesterday~~** have been given by more boring professors **than** λm. **I met** m **~~professors~~** **yesterday**. (22b)

If *meet* in (27) is replaced by *evaluate*, a verb compatible with both *lectures* and *professors*, the resulting sentence in (28), which is parallel to (27), is acceptable.

(28) Context: As chair of the academic committee, I evaluate courses and professors annually. This year, teaching quality has declined significantly, with increases in both silly lectures and boring professors.
More silly lectures than I evaluated last year have been given by more boring professors than I evaluated last year.

[6] (25) involves a conjunction of comparative expressions. I will discuss this in §5.3.

Under the reconstruction analysis, (29) would be analyzed as (28) and therefore be equally acceptable. But this prediction is not borne out: (29) is intuitively as weird as (22b). The contrast between (29) and (28) argues against the reconstruction analysis.

(29) *<u>More</u> silly lectures have been given by <u>more</u> boring professors than I evaluated last year.

Taken together, the discussions of Chomsky (1981/1993) and von Stechow (1984) show that the syntax of *than*-clauses in MHCs should be no different from that of comparatives in general. Chomsky (1981/1993)'s observations on comparatives motivate a general analysis involving (i) **ellipsis** and (ii) **anaphoric dependency** exempt from subjacency, making von Stechow (1984)'s MHC-specific reconstruction of covert *than*-clauses unnecessary. Accordingly, as illustrated in (30) and (31), the *than*-clauses of comparatives (including MHCs) admit a uniform syntactic analysis.

(30) [More silly lectures] have been given by [more boring professors] . . .

a. (22a): than [$\lambda m.\lambda n.$ I would have expected ~~m-many silly lectures have been given by n-many boring professors~~] **Multiple elided gradable adjectives**

b. (22b): than [$\lambda n.$ I met ~~n-many boring professors~~ yesterday]

(31) Pictures of [more people] are for sale . . .

a. (23a): than [$\lambda n.$ I expected ~~pictures of n-many people are for sale~~]

b. (23b): than [$\lambda n.$ I met ~~n-many people~~ yesterday]

Thus, MHCs require no syntactic machinery beyond that already needed for comparatives in general. The degradedness of (22b) is therefore not unique to MHCs but is shared by comparatives like (23b), suggesting an explanation outside syntax.

2.3.2 Semantic composition

Hendriks (1994) argues that von Stechow (1984)'s 'two independent comparisons' analysis in (10) over-generates. She questions the acceptability of MHC examples like (32) and proposes the analysis in (33) (see Hendriks 1994: (8)).

(32) %<u>More</u> dogs ate <u>more</u> rats than cats ate mice. (= (7), von Stechow 1984: (140))

(33) the number of dogs x such that '|rats eaten by x| > |mice eaten by y|' > the number of cats y such that '|rats eaten by x| > |mice eaten by y|'

Under Hendriks (1994)'s analysis, the two comparisons are not independent (cf. (10)): in (33), one comparison is embedded within another. The inner comparison '|rats eaten by x| > |mice eaten by y|' restricts the items x and y, whose cardinalities are then compared. Hendriks (1994) claims that this restriction '|rats eaten by x| > |mice eaten by y|' makes x and y mutually dependent and gives rise to infinite regress: the definition of x relies on y, and vice versa. Thus, this mutual dependency creates a problem for semantic composition, leading to the degradedness of (32).

However, many MHCs are fully acceptable and readily interpretable, suggesting that their semantic derivation does not suffer from infinite regress. Thus the degradedness of examples like (32) needs a different explanation.

In §3 and §4, I build on Brasoveanu (2013)'s **dynamic-semantic** analysis of cumulative-reading sentences to show that the mutual restriction described in §2.1 is compatible with semantic compositionality.[7]

2.3.3 Pragmatic account

The discussions in §2.3.1 and §2.3.2 suggest that the degradedness of MHCs such as (22b) and (32) cannot be attributed to syntactic structure or semantic composition. This is consistent with Meier (2001)'s remark that 'the fact that (34b) is somehow odd is just a reflex of the fact that it is hard to imagine a situation in which the comparison construction might be relevant (p. 356)'. The same observation naturally extends to all the examples in (34), all of which are judged degraded by Hendriks (1994).

(34) a. %<u>More</u> dogs ate <u>more</u> rats than cats ate mice. (=(7)/(32))
b. ??Fewer dogs ate more rats than cats ate mice. (Meier 2001: p. 356, (17a))
c. ?More doors are higher than windows are wide. (Meier 2001: p. 356, (17b))

Examples in (35) share the same syntax and compositional mechanisms as those in (34), yet remain fully natural and interpretable. Thus the contrast points to a **pragmatic** account. In (35a), the two increases jointly signal a deterioration in teaching quality; in (35b), the decrease and the increase together reflect improved corn productivity; and in (35c), the two changes together indicate a technological advance. The generalization is that these MHCs all involve two changes that are meaningfully related.

[7] More broadly, beyond the literature on cumulative-reading sentences, Bumford (2017) investigates the mutual definition of uniqueness in Haddock descriptions (e.g., *the rabbit in the hat*, see Haddock 1987) and proposes an analysis in the same spirit as Brasoveanu (2013). See also Charlow (2017).

(35) a. More silly lectures have been given by more boring professors – than I would have expected. (=(1))

b. Less land produces more corn than ever before. (=(2))

c. Newer generations of microchips contain more electronic switches on a smaller surface. (=(18))

As illustrated in (36), once the context makes it salient that the two changes interact in addressing an underlying issue (e.g., the economics of keeping animals), a sentence with the same structure as (34b) becomes much more interpretable.[8]

(36) Context: A zoo keeper is comparing the cost of feeding different animals. The zoo keeps fewer lions than chimpanzees, but lions eat more food overall, and the meat they consume is much more expensive than the fruit eaten by chimpanzees.
Fewer lions eat more meat than chimpanzees eat fruit. (cf. (34b))

This pragmatic requirement on MHC interpretability echoes existing observations that comparison is not merely a matter of ordering degrees. Rather, comparison is often context-sensitive, drawing on parallelism between the alternatives under comparison to serve a conversational goal (see, e.g., Li 2023). The acceptability gradient in (37) illustrates that the felicity of a comparison depends on the salience of an underlying issue, as reflected in the relevance between the compared alternatives, with parallelism enhancing relevance. In other words, comparison is felicitous only when the relevant increase or decrease advances the conversational goal, rather than merely reflecting arbitrary differences between unrelated degrees.[9]

(37) a. He met more celebrities in this restaurant today than I met yesterday.

b. More celebrities came to this restaurant than I met yesterday.

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[8] I thank Takeo Kurafuji for discussing example (34b) with me and providing this scenario in (36).

[9] More broadly, the current pragmatic view also explains other observations in the literature. Beck et al. (2012) note that (ia) is more acceptable than (ib), while this contrast between *expensive* and *long* disappears in (ic), where adjectival ellipsis licenses the recovery of either adjective. Presumably, without adjectival ellipsis, price comparison is more salient than length comparison, making (ia) easier to interpret than (ib). Length comparison becomes salient in (ii), where the comparison is explicitly restricted to umbrellas.

(i) a. Compared to what John bought, Mary bought a more expensive umbrella.

b. *Compared to what John bought, Mary bought a longer umbrella.

c. Mary bought a more expensive / a longer umbrella than John did.
[than λd. John did ~~a d-expensive/long umbrella~~]. (Ellipsis is compatible with either adjective.)

(ii) Compared to the umbrella John bought, Mary bought a longer one.

c. Pictures of more singers were displayed than gave a performance yesterday.

d. *Pictures of more people are for sale than I met yesterday. (=(23b))

Therefore, for an MHC to be interpretable, it pragmatically requires (i) an interplay between the multi-dimensional changes, and (ii) relevance (i.e., parallelism) between the matrix clause and the *than*-clause. The degraded examples discussed in §2.3.1 (repeated in (38) and (39)) fail to satisfy these requirements, particularly the second.

(38) *[More silly lectures] have been given by [more boring professors] – than I met yesterday. (=(22b))

(39) *A greater man would be a better man than Otto. (=(24))

It is also possible for a comparative to contain multiple comparative expressions while lacking a contextually salient issue that would license an interaction between the multi-dimensional changes, as illustrated in (40)–(42) and the Portuguese example (43):

(40) More schools choose a *longer* school year than before. (= (8))
$\rightsquigarrow$ than [λn. ~~n-many schools chose a longer school year~~ before]

(41) John made more people *prettier* than I thought he would. (Hendriks 1994: (12))
$\rightsquigarrow$ than [λn. I thought he would ~~make n-many people prettier~~]

(42) (The world would be a more positive place to be if) more people spent *more* of their time focusing on what they're thankful for.
(https://schizanthusnerd.com/2018/04/01/everyday-gratitude/)
$\rightsquigarrow$ ~~than [λn. n-many people spent more of their time focusing on ... now]~~

(43) Eles querem que mais empresas paguem *menos* IRC
they want that more companies pay less corporate-tax
'They want more enterprises to pay less corporate tax.' (Marques 2005: (65))
$\rightsquigarrow$ ~~than [λn. n-many enterprises pay less corporate tax currently]~~

Given our world knowledge, the most natural interpretation of (40)–(43) involves only an increase in the number of schools, people, or enterprises that each undergoes the relevant change. Consequently, one of the comparative expressions (the italicized one) merely restricts the schools, people, or enterprises under comparison. This interpretation is further evidenced by the preferred intonation pattern: speakers tend to stress only the underlined *more*, indicating which alternatives are relevant for comparison.

2.4 Interim summary

To sum up, MHCs display variants comparable to those of regular comparatives (§2.2). MHCs share the same syntax and semantics of regular comparatives (§2.3.1 and §2.3.2): they involve (i) ellipsis in the *than*-clause and (ii) an anaphoric dependency between comparative expressions and their standards of comparison. Pragmatically (§2.3.3), MHCs also pattern with regular comparatives in using comparison to address an underlying issue and thus fulfil the conversational goal.

What distinguishes MHCs is that there are multi-dimensional changes, so that (i) they interact to jointly address a single conversational goal, and (ii) the same interaction is manifested in parallel in the matrix clause and the *than*-clause.

The next section turns to cumulative-reading sentences to further examine multi-dimensional interaction.

3 Inspiration from cumulative-reading sentences

§2 shows that MHC interpretation involves an interplay between multi-dimensional changes, which reflects a contextually salient issue. This section shows that this multi-dimensional interplay also underlies the meaning of cumulative-reading sentences. I first review the compositional mechanism proposed by Brasoveanu (2013) (§3.1). Then based on Krifka (1999), I show how the underlying issue and the multi-dimensional interplay affect sentence interpretation and compositionality. Building on these insights, §3.3 develops an informal analysis of MHCs.[10]

3.1 Brasoveanu (2013)'s analysis of cumulative-reading sentences

A sentence like (44) contains two **modified numerals** (see the underlined parts) and has a **cumulative** reading:[11]

(44) <u>Exactly three boys</u> saw <u>exactly five movies</u>. (see e.g., Brasoveanu 2013)
Cumulative: Some boys saw some movies. The total number of boys who saw any movies is 3, and the total number of movies seen by any boys is 5.

[10] Cumulative-reading sentences express multiple quantity measurements, but MHCs are not limited to quantity comparisons (see §2.2). §4 extends the analysis to MHCs beyond quantity comparisons (e.g., (18)).

[11] Sentence (44) also has a **distributive** reading: There are exactly 3 boys such that each of them saw exactly 5 movies. I do not discuss this reading in this paper. See Brasoveanu (2013) for details.

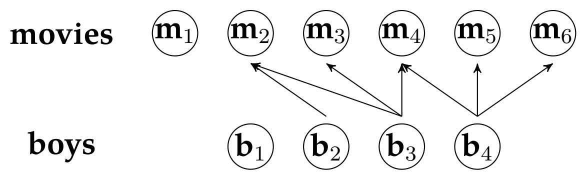

Figure 1: The genuine **cumulative** reading of (44) is **true** in this context.

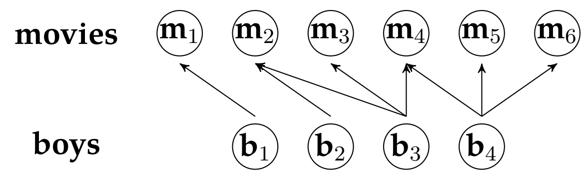

Figure 2: The genuine **cumulative** reading of (44) is **false** in this context.

Krifka (1999), Landman (2000), and Brasoveanu (2013) note that the derivation of this cumulative reading is based on a **mutual restriction** between the two modified numerals, so that they are interpreted simultaneously and no scope-taking is involved.

As shown in (45), if one modified numeral (here *exactly three boys*) takes scope over the other (here *exactly five movies*), then only the latter is involved in restricting the former, but not vice versa. This would yield the pseudo-cumulative reading:

(45) **Pseudo-cumulative reading** of (44): The maximal boy-sum such that 'they saw five movies in total' has a cardinality of 3.
$\rightsquigarrow$ True in Fig. 2: there are two boy-sum witnesses, $\mathbf{b}_2 \oplus \mathbf{b}_3 \oplus \mathbf{b}_4$ and $\mathbf{b}_1 \oplus \mathbf{b}_2 \oplus \mathbf{b}_4$. No larger boy-sum satisfies the restriction that 'they saw five movies in total'.

The reading in (45) is true in Fig. 2, but sentence (44) is intuitively false in Fig. 2, indicating that this pseudo-cumulative reading in (45) is too weak and not readily available. Thus the genuine cumulative reading of (44) must involve mutual restriction.[12]

Under the scenario in Fig. 1, the genuine cumulative reading (see (44)) is true. Here, *exactly three boys* counts the totality of **boys who saw any movies**, which is 3 (cf. the cardinality of all boys in the context is 4), and *exactly five movies* counts the totality of **movies seen by any boys**, which is 5 (cf. the cardinality of all movies in the context is 6).

Brasoveanu (2013) adopts **dynamic semantics** to implement these ideas and develops a compositional analysis of the cumulative reading of (44). Modified numerals make semantic contributions in several layers, as sketched in (46):

(46) Exactly three^x boys saw exactly five^y movies. **Cumulative reading** of (44)

$$\underbrace{\sigma x \sigma y[\textsc{boy}(x) \wedge \textsc{movie}(y) \wedge \textsc{saw}(x,y)]}_{\text{the mereologically maximal } x \text{ and } y} \wedge \underbrace{|y| = 5 \wedge |x| = 3}_{\textbf{post-suppositional} \text{ cardinality tests on selected drefs}}$$

[12]In fact, a reading à la (45) can be licensed by an appropriate QUD (see Buccola and Spector 2016, Zhang 2018): Consider, in Fig. 2, *how many boys saw exactly 5 movies between them?* Further constraints apply (see e.g., Gentile and Schwarz 2018, 2020), and I do not pursue this issue here.

First, each modified numeral introduces a potentially plural **discourse referent** (**dref**). Then restrictions are added onto drefs: here $\textsc{boy}(x)$, $\textsc{movie}(y)$, and $\textsc{saw}(x, y)$.[13]

Second, after all the drefs are introduced and restrictions are added, modified numerals contribute **mereology-based maximality operators** – σ (similar to a Link-style sum operator). These maximality operators are applied at the sentence level simultaneously, selecting the maximal drefs satisfying the restrictions.

Finally, the modified numerals contribute **post-suppositional cardinality tests**, checking whether the cardinality of the relativized maximal boy-sum (i.e., those who saw movies) is 3 and whether the cardinality of the relativized maximal movie-sum is 5. According to Brasoveanu (2013), these cardinality tests are post-suppositional in the sense that they are applied on the output context, i.e., after maximization, as opposed to presuppositions, which are tests on the input context.[14]

Evidently, cumulative-reading sentences, which express multi-dimensional measurements, and MHCs, which express multi-dimensional comparisons, are closely parallel. As shown in (47), the interpretation of this MHC also relies on mutual restriction between professors and lectures, in both the matrix clause and the *than*-clause. Mutual restriction leads to relativized maximization, excluding silly lectures taught by graduate students and boring professors who did not teach. Just as post-suppositional cardinality tests in (46) are applied after maximization, the comparisons in (47) must be performed on the relativized maximal sums of professors and lectures. In other words, comparisons function like post-suppositional tests.[15]

(47) More^{y} silly lectures have been given by more^{x} boring professors than [$\lambda m.\lambda n.$ I would have expected ~~m-many$^{y'}$ silly lectures have been given by n-many$^{x'}$ boring professors~~] (=(1)/(10), see §2.1)

$$\underbrace{\sigma x \sigma y[\textsc{b-prof}(x) \wedge \textsc{s-lecture}(y) \wedge \textsc{give}(x, y)]}_{\text{the mereologically maximal } x \text{ and } y \text{ in the matrix clause}} \wedge$$

$$\underbrace{\sigma x' \sigma y'[\textsc{b-prof}(x') \wedge \textsc{s-lecture}(y') \wedge \textsc{give}(x', y')]}_{\text{the mereologically maximal } x' \text{ and } y' \text{ in the } \textit{than}\text{-clause}} \wedge \underbrace{|y| > |y'| \wedge |x| > |x'|}_{\textbf{post-suppositional} \text{ comparison tests on selected drefs}}$$

[13]Cumulative closure is assumed for lexical relations when needed.

[14]Haslinger and Schmitt (2020)'s recent analysis of cumulative readings of modified numerals is based on similar ideas: maximization contributed by modified numerals is not local, but takes the widest scope, i.e., at the sentential level. They also note that cumulative readings still exhibit structural asymmetry between modified numerals, but this issue is orthogonal to the current paper.

[15]In (47), *more* is the comparative form of *many*, corresponding to the gradable adjectives elided in the *than*-clause.

Then how to understand the simultaneous application of these maximality operators σ? Is simultaneous application a stipulation? Must maximization be mereology-based? How does (47) differ from von Stechow (1984)'s analysis in (10)? §3.2 addresses these questions and shows that maximization, which plays a crucial role in dref selection, is driven by interlocutors' underlying concern, i.e., a QUD.

3.2 Interplay between multi-dimensional measurements

Krifka (1999) discusses cumulative-reading sentences like (48), questioning the **simultaneous mereology-based maximization** strategy used in (46):

(48) In Guatemala, at most 3% of the population own at least 70% of the land.

Krifka (1999) points out that

> '…Under the simplifying (and wrong) assumption that foreigners do not own land in Guatemala, and all the land of Guatemala is owned by someone, this strategy would lead us to select the alternative *In Guatemala, 100 percent of the population own 100 percent of the land*, which clearly is not the most informative one among the alternatives — as a matter of fact, it is pretty uninformative. We cannot blame this on the fact that the NPs in (48) refer to percentages, as we could equally well express a similar statistical generalization with the following sentence (assume that Guatemala has 10 million inhabitants and has an area of 100,000 square kilometers):
>
> *In Guatemala, 300,000 inhabitants own 70,000 square kilometers of land*.
>
> 'Again, the alternative *In Guatemala, 10 million inhabitants own 100,000 square kilometers of land* would be uninformative, under the background assumptions given.
>
> 'What is peculiar with sentences like (48) is that they want to give information about the bias of a statistical distribution. One conventionalized way of expressing particularly biased distributions is to select a small set among one dimension that is related to a large set of the other dimension. Obviously, to characterize the distribution correctly, one should try to decrease the first set, and increase the second. In terms of informativity of propositions, if (48) is true, then there will be alternative true sentences of the form *In Guatemala*, n *percent of the population own* m *percent of the land*, where n

is greater than *three*, and m is smaller than *seventy* (see Fig. 4 in the current paper). But these alternatives will not entail (48), and they will give a less accurate picture of the skewing of the land distribution.' (Krifka 1999: §3.1)

Haslinger and Schmitt (2020) also note that a sentence like (48) has a non-maximal cumulative reading, and argue that 'non-maximal readings of modified numerals can only occur under a certain condition, which involves a "significantly more surprising" relation that is probably related to the semantics of scalar particles like *even*,' and that this relation 'must be context-dependent. (p. 335)'

These discussions suggest that although the cumulative reading of (48) is '**non-maximal**', it is highly informative along a contextually relevant scale, reminiscent of *even*-sentences, expressing a 'particularly biased distribution'.[16]

More generally, these observations show that maximal informativeness does not necessarily coincide with mereological maximality (or maximal values along each scale). Rather, as Beck and Rullmann (1999) already observe (see also Beck 2013, von Fintel et al. 2014, a.o.), maximal informativeness depends on the **degree QUD** involved, and in particular on its **predicate**, which induces different patterns of entailment.

With an **upward monotone** degree predicate P as in (49), a higher degree corresponds to higher informativeness. Thus maximal informativeness is achieved by the dref x with the maximal cardinality (i.e., the sum of all books Mary read), and its cardinality information (here $|x| < 5$) resolves the QUD. In contrast, with a **downward monotone** degree predicate P as in (50), a lower degree corresponds to higher informativeness. Thus maximal informativeness is achieved by the dref x with the minimal cardinality, and its cardinality information resolves the QUD (here $|x| \approx 5$).

(49) Mary read <u>fewer than 5^x</u> books. ($P = \lambda n.$ Mary read n-many books)
P is **upward monotone**: $\forall m, n[m \geq n \rightarrow P(m) \vDash P(n)]$

(50) <u>About 5^x</u> eggs are sufficient for this. ($P = \lambda n.$ n-many eggs are sufficient for this)
P is **downward monotone**: $\forall m, n[m \leq n \rightarrow P(m) \vDash P(n)]$

[16] Haslinger and Schmitt (2020) note that a sentence like (48) has a 'surprising effect' (i.e., mirativity, norm-sensitivity). According to Krifka (2000), along a contextually relevant scale, our expected value can be regarded as the average of the relevant alternatives. Since minimal and maximal values necessarily depart from this expectation, they naturally give rise to mirativity. This explains the mirativity associated with *even*-like particles, whose prejacent is maximally informative among the relevant alternatives on a contextually relevant scale (see Greenberg 2018, Zhang 2022). The intuitive mirativity of (48) therefore provides further evidence that its 'non-maximal' cumulative reading is based on a scale and involves maximal informativeness. See also §5.1 on *even* and footnote 20 on norm-sensitivity.

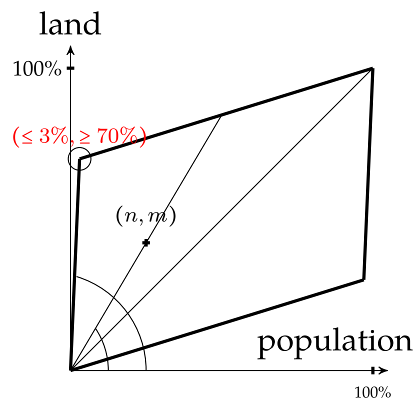

Figure 3: (48): *At most 3%^{x} of the population own at least 70%^{y} of the land*.
The percentage measurements of the drefs, ($\textsc{pct}(x), \textsc{pct}(y)$), form a parallelogram in the measurement space. Maximal informativeness is achieved at the top-left corner, whose measurements are reported: ($\leq 3\%, \geq 70\%$).

The above understanding of QUD-based informativeness extends naturally from one-dimensional to multi-dimensional measurement.

With a **small** value along one dimension ($\leq 3\%$ *of the population*) contrasted with a **large** value along the other ($\geq 70\%$ *of the land*), the **non-maximal** cumulative reading of (48) addresses a degree QUD along a **composite scale of unevenness**. Maximal informativeness is achieved by the drefs whose measurements jointly maximize unevenness (see the top-left corner in Fig. 4, where, e.g., $\frac{y/x}{(1-y)/(1-x)}$ is maximal, indicating the greatest disparity in land owned per capita between the selected population and the remaining population). The measurements of the selected drefs resolve the QUD.

The precise degree QUD cannot be uniquely recovered from (48) alone. (48) may address *how unbalanced the wealth distribution is*, *how concentrated land ownership is*, *how severe inequality is*, etc. Thus it is compatible with a family of degree QUDs that involve composite scales of unevenness based on population and land-ownership measures.

Unlike the **non-maximal** cumulative reading of (48), **maximal** cumulative readings are based on the selection of drefs that are maximal along each dimension (see the top-right corners of Figs. 4 and 5), and there are two classes of QUDs.

As illustrated in (51) (see Fig. 4), when the two dimensions contribute **in parallel** to a composite scale, the cumulative reading addresses **overall magnitude**. For example, (51) is compatible with degree QUDs such as *How bad is the teaching quality?*, *How disappointing is higher education?*, etc. Under such QUDs, maximal informativeness is

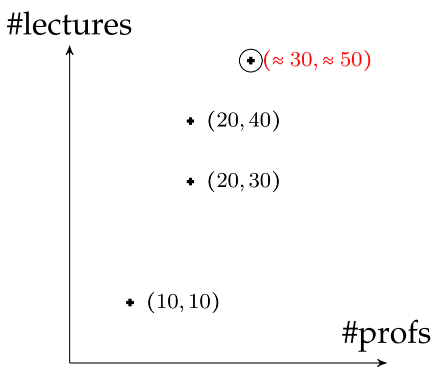

Figure 4: (51): *About 50 silly lectures were given by about 30 boring professors.*

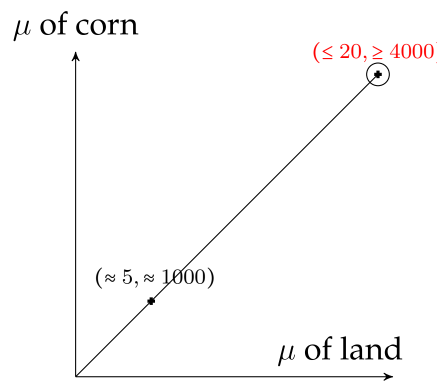

Figure 5: (52): *Less than 20 acres yielded over 4,000 bushels of corn.*

naturally achieved by jointly maximizing the cardinalities of lectures and professors.

Brasoveanu (2013)'s example in (44) can be interpreted in the same way. It is compatible with degree QUDs such as *How much movie consumption is there among boys?*, *How poor is the movie market among boys?*, etc. In this sense, the simultaneous application of mereology-based maximality operators is not a stipulation in (46), but rather driven by the implicit assumption of such QUDs.

By contrast, as illustrated in (52) (see Fig. 5), the two dimensions contribute to a composite scale **in opposite directions**, and the cumulative reading addresses a degree QUD whose underlying composite scale is a **constant rate**: e.g., corn productivity in the interpretation of (52). Accordingly, all relevant drefs lie on the same line in Fig. 5, sharing the same value on the composite productivity scale. Maximal informativeness can be modeled by a lexicographic ordering, e.g., $G(\langle x, y\rangle) = (\frac{y}{x}, y)$, which first ranks drefs by productivity $\frac{y}{x}$ and, among drefs with equal productivity, selects the mereologically largest land and corn drefs (see the top-right corner of Fig. 5). The measurements of these selected drefs then resolve the underlying degree QUD.

(51) About 50 silly lectures were given by about 30 boring professors. (see Fig. 4)
$\rightsquigarrow$ *How bad was the teaching quality?* (Joint reinforcement of deterioration)
$\rightsquigarrow$ **So many** silly lectures were given by **so many** boring professors.

(52) Less than 20 acres yielded over 4,000 bushels of corn. (see Fig. 5)
$\rightsquigarrow$ *How productive is this land?* (A constant rate between the two dimensions)
$\rightsquigarrow$ **So little** farmland yielded **so much** corn.

As summarized in Table (53), the discussion above identifies three classes of cumulative readings, which differ in how multiple dimensions contribute to the composite scale underlying the relevant degree QUDs (see the third and fourth columns). When interpreting a cumulative-reading sentence, the precise degree QUD is context-dependent, but the class of reading is often suggested by lexical cues. For example, percentages are a reliable indicator of the non-maximal reading, whereas the monotonicity of modified numerals suggests whether the multiple dimensions contribute to the composite scale in parallel or in opposite directions. In (48), for instance, *at most 3%* is downward-entailing, whereas *at least 70%* is upward-entailing.

(53) **Three classes of cumulative readings**

Reading	Potential lexical clues	Composite scale	Composite-scale function (and monotonicity)
Non-maximal (Fig. 3)	Percentages; opposite-monotonicity modified numerals	e.g., unevenness, disparity	e.g., $G(\langle x, y\rangle) = \frac{y/x}{(1-y)/(1-x)}$ ($\uparrow y, \downarrow x \Rightarrow \uparrow G$)
Maximal (Fig. 4)	Parallel-monotonicity modified numerals	overall magnitude	e.g., $G(\langle x, y\rangle) = \langle x, y\rangle$ (coordinatewise order) ($\uparrow y, \uparrow x \Rightarrow \uparrow G$)
Maximal (Fig. 5)	Opposite-monotonicity modified numerals	e.g., productivity, efficiency, value for money	e.g., $G(\langle x, y\rangle) = \langle \frac{y}{x}, y\rangle$ (lexicographic order) ($\uparrow \frac{y}{x} \Rightarrow \uparrow G$; with fixed $\frac{y}{x}, \uparrow y \Rightarrow \uparrow G$)

Sometimes, a sentence can be ambiguous between two classes of cumulative readings, as illustrated in (54).

(54) (Fewer than) 300,000 inhabitants own (more than) 70,000 km^2 of land.

a. **Non-maximal**: A scale of unevenness $\rightsquigarrow$ the wealth distribution is skewed

b. **Maximal**: A scale of per-capita wealth $\rightsquigarrow$ these people are wealthy

To sum up, cumulative-reading sentences use multiple dimensions to address a single degree QUD defined over a composite scale. The different classes of cumulative readings differ in how maximal informativeness on this composite scale is related to the underlying multi-dimensional measurements, which in turn determines which drefs are selected. This QUD-based perspective on cumulative readings paves the way for the analysis of MHCs.

3.3 From multi-measurement sentences to MHCs

An MHC extends multi-dimensional measurement to multi-dimensional comparison. Corresponding to the three classes of cumulative readings summarized in (53), three classes of MHCs can be distinguished, each addressing a degree QUD along a different class of composite scale, as summarized in (55):

(I) a composite scale of **unevenness**, corresponding to the non-maximal, percentage-based cumulative reading (see (56) and Fig. 6);

(II) a composite scale of **overall magnitude**, corresponding to the maximal, parallelism-based cumulative reading (see (57) and Fig. 7);

(III) a composite scale of **constant rate**, corresponding to the maximal, constant-rate-based cumulative reading (see (58) and (59), and Figs. 8 and 9).

(55) **Three classes of MHCs**

Examples	Along the composite scale	Dimensions of comparison
The percentage reading of (56)	Increase in unevenness (see $\beta > \alpha$ in Fig. 6)	population ↘ vs. wealth ↗ (opposite comparative expressions)
(57) (see also (3), (6))	Deterioration in teaching quality (larger distance to the zero point in Fig. 7)	lectures ↗ and profs ↗ (parallel comparative expressions)
(58) and (59) (see also (18), (36))	Increase in productivity;	land ↘ vs. corn ↗
	Decrease in value for money (see $\beta > \alpha$ in Fig. 8 and 9)	A small increase of quality vs. a large increase of price

(56) Fewer^x people own more^y wealth here than [$\lambda m.\lambda n.$ ~~m-many$^{x'}$ people own n-much$^{y'}$ wealth~~ elsewhere]. (The non-maximal, percentage reading: see Fig. 6)
$\rightsquigarrow$ Along the composite scale of **unevenness**: here vs. elsewhere

(57) More^y silly lectures have been given by more^x boring professors than [$\lambda m.\lambda n.$ I would have expected ~~m-many$^{y'}$ silly lectures have been given by n-many$^{x'}$ boring professors~~]. (see Fig. 7)
$\rightsquigarrow$ Along the composite scale of **teaching quality**: reality vs. expectation

(58) Less^x land produces more^y corn than [$\lambda m.\lambda n.$ ~~m-much$^{x'}$ land produced n-much$^{y'}$ corn~~ ever before]. (The maximal, non-percentage reading: see Fig. 8)
$\rightsquigarrow$ Along the composite scale of **corn productivity**: now vs. before

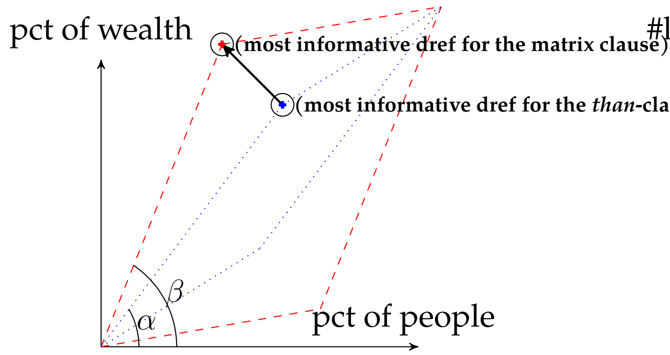

Figure 6: (56): *Fewer people own more wealth here than elsewhere.*

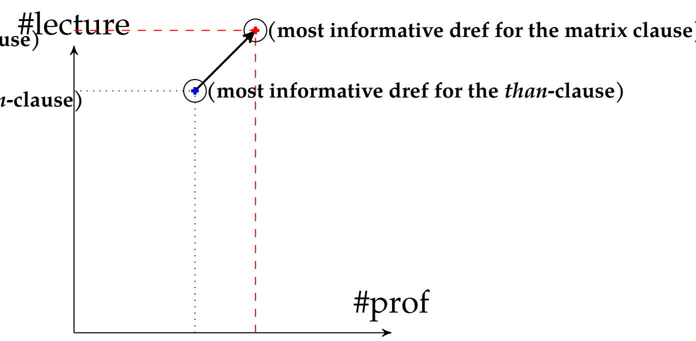

Figure 7: (57): *More silly lectures have been given by more boring professors than I would have expected.*

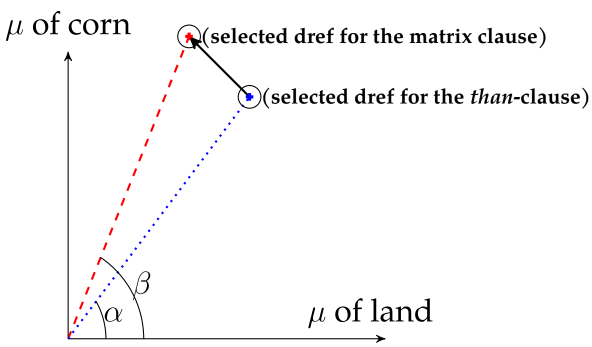

Figure 8: (58): *Less land produces more corn than ever before.*

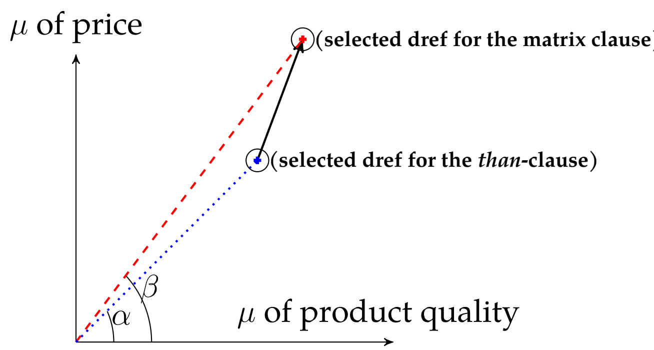

Figure 9: (59): *Slightly better products cost a lot more than last year.*

(59) Slightly better^x products cost a lot more^y than [$\lambda m.\lambda n.$ ~~m-good$^{x'}$ products cost n-much$^{y'}$~~ last year]. (see Fig. 9)
(adapted from https://shavercheck.com/best-electric-razor/)
$\rightsquigarrow$ Along the composite scale of **value for money**: now vs. last year[17]

In regular, single-dimensional comparatives, the same scale (i.e., the one associated with the comparative expression) both determines maximal informativeness for dref selection and serves as the scale for dref comparison:

(60) a. Lucy is {taller / less tall} than [λd. Mary is ~~d-tall~~]. Tallness
b. Mary is shorter than [λd. Lucy is ~~d-short~~]. Shortness

[17]Unlike cumulative readings, which involve two quantity measurements, this MHC in (59) combines a quantity and a non-quantity measurement. The parallelism between the two constructions concerns their shared degree-QUD-based interpretation mechanism, not their lexical scales.

However, MHCs pattern with cumulative-reading sentences in distinguishing between two kinds of scales. First, a **hidden, QUD-driven composite scale** determines maximal informativeness and thereby selects the relevant drefs in both the matrix and *than*-clauses. Second, the selected drefs are measured or compared along **the multiple dimensions overtly expressed** by the modified numerals or comparative expressions.

In this interpretation strategy, lexical information conveyed locally by modified numerals or comparative expressions (including scales, as well as monotonicity or polarity, which indicate parallelism or contrast and information projection patterns) first signals a global conversational goal, i.e., a degree QUD along a composite scale. This QUD then guides sentence interpretation and feeds back into local semantic computation (by selecting drefs for post-suppositional tests).

This interaction between local lexical cues and global discourse structure is found more generally in natural language (see also §5.2). The lexical information conveyed by modified numerals and comparative expressions plays a role similar to prosodic focus in (61), which signals a relevant QUD and thus determines the most informative dref (here, the maximal sum of the individuals Lucy introduced to Sue). A post-suppositional test then checks whether 'Mary $\oplus$ Naomi' is part of this dref.

(61) Lucy introduced [Mary and Naomi]$_F$ to Sue. $\rightsquigarrow$ Who did Lucy introduce to Sue?

This proposed analysis improves on von Stechow's (1984) analysis in (10) (see §2.1) in two respects. First, by distinguishing (i) the hidden QUD-driven composite scale responsible for dref selection from (ii) the overt dimensions along which the selected drefs are compared, it naturally acounts for MHCs that express an increase in unevenness (see (56) and Fig. 6). By contrast, von Stechow (1984)'s analysis would compare the entire population and the total wealth of the two places (i.e., 100% vs. 100% in both dimensions), thereby missing the the increase-in-unevenness reading of (56).

Second, the connection to cumulative-reading sentences and the notion of a hidden composite scale provide a precise and exhaustive classification of MHC interpretations. Parallel comparative expressions signal a change in overall magnitude (see Fig. 7), whereas contrasts across dimensions signal changes in unevenness or in a constant rate (see Figs. 6, 8, and 9). By taking into account how multiple dimensions jointly contribute to the underlying conversational goal, the present proposal avoids the over-generation problems identified by Hendriks (1994).

The next section develops a formal implementation of this proposal.

4 Formal analysis of multi-head comparatives

I first develop a dynamic semantic analysis of maximal informativeness for cumulative-reading sentences (§4.1), then recast comparatives within this framework (§4.2), and finally combine the two to analyze MHCs (§4.3).

4.1 The meaning of a cumulative-reading sentence

Following Brasoveanu (2013), I derive the meaning of cumulative-reading sentences within Dynamic Predicate Logic (DPL, Groenendijk and Stokhof 1991). A dynamic framework naturally facilitates the introduction of and subsequent operations on drefs.

Within dynamic semantics, meaning derivation is considered a series of updates from an information state to another. Given the distributivity of DPL (see (62b), Groenendijk and Stokhof 1990), an update is a relation between assignment functions (i.e., it takes an assignment function as input and returns a set of assignment functions).

(62) DPL (see Groenendijk and Stokhof 1990, 1991):

a. **Information state** i: a set of assignment functions g — Type: $\langle st \rangle$

b. **Update** r: from an information state to another — Type: $\langle st, st \rangle$
Given that for every information state i, $r(i) = \bigcup_{g \in i} r(\{g\})$, an update can be considered a relation between assignment functions (of type $\langle s, st \rangle$).

c. **Truth**: an update is true if it yields a non-empty set.

Modified numerals make semantic contributions in several layers (see Brasoveanu 2013 and §3.1): (i) introducing drefs; (ii) imposing a degree-QUD-based maximality operator $\mathbf{M}_{\text{QUD-INFO}}$ for dref selection; (iii) imposing cardinality tests on the selected drefs.

As shown in (63), $\mathbf{M}_{\text{QUD-INFO}}$ takes an input update m and returns an update such that the assigned drefs (e.g., $h(u_i), h(u_j), \ldots$) maximize informativeness with respect to the degree-QUD-driven composite scale, G_{QUD}.

G_{QUD} is defined over measure functions $\mu_i, \mu_j \ldots$, which measure drefs along dimensions such as cardinality, percentage, etc. G_{QUD} takes a tuple of measurements of drefs and returns either a real number or an ordered pair: their ordering determines the informativeness of drefs along the composite scale associated with the relevant QUD.

(63) $$\mathbf{M}_{\text{QUD-INFO}} \stackrel{\text{def}}{=} \lambda m_{\langle s,st \rangle}.\lambda g_s.\{h \in m(g) \mid \neg\exists h' \in m(g). \\ G_{\text{QUD}}(\langle \mu_1(h'(u_1)), \mu_2(h'(u_2)), \ldots \rangle) \succ_{\text{INFO}} G_{\text{QUD}}(\langle \mu_1(h(u_1)), \mu_2(h(u_2)), \ldots \rangle)\}$$

The three classes of cumulative readings discussed in §3.2 (see (53)) correspond to three classes of composite scales, represented by G_{uneven}, G_{mag}, and G_{rate}. As defined in (64)–(66), G_{uneven} captures the non-maximal cumulative reading associated with **unevenness**, while G_{mag} and G_{rate} capture maximal cumulative readings associated with **overall magnitude** and **constant rate**, respectively.

(64) $G_{\text{uneven}}(\langle\%(x), \%(y)\rangle) \stackrel{\text{def}}{=} \frac{\%(y)/\%(x)}{(1-\%(y))/(1-\%(x))}$ (e.g., the ratio of $\frac{\%(y)}{\%(x)}$ to $\frac{1-\%(y)}{1-\%(x)}$)

(65) $G_{\text{mag}}(\langle x, y\rangle) \stackrel{\text{def}}{=} \langle x, y\rangle$, where $G_{\text{mag}}(\langle x_1, y_1\rangle) \geq_{\text{coord}} G_{\text{mag}}(\langle x_2, y_2\rangle)$ iff $x_1 \geq x_2 \wedge y_1 \geq y_2$

(66) $G_{\text{rate}}(\langle x, y\rangle) \stackrel{\text{def}}{=} \langle \frac{y}{x}, y\rangle$, where $G_{\text{rate}}(\langle x_1, y_1\rangle) \geq_{\text{lex}} G_{\text{rate}}(\langle x_2, y_2\rangle)$ iff $\frac{y_1}{x_1} > \frac{y_2}{x_2}$ or $\frac{y_1}{x_1} = \frac{y_2}{x_2} \wedge y_1 > y_2$

Based on G_{uneven} and the cardinality tests in (68), (67) derives the non-maximal cumulative reading step by step.

(67) $[\![$At most 3%^{u} of the population own at least 70%$^\nu$ of the land$]\!]$ (=(48))

$$\Leftrightarrow \underbrace{\textbf{at-most-3\%}_u[\textbf{at-least-70\%}_\nu}_{\text{tests}}[\underbrace{\mathbf{M}_{\text{QUD-INFO}}}_{\text{dref selection}}\ \underbrace{[\![\text{some}^u \text{ people own some}^\nu \text{ land}]\!]}_{\text{dref introduction}}]]$$

a. $\Leftrightarrow$

$$\textbf{at-most-3\%}_u[\textbf{at-least-70\%}_\nu[\mathbf{M}_{\text{QUD-INFO}}[\lambda g.\ \left\{g^{u\mapsto x}_{\nu\mapsto y} \middle| \ \textsc{land}(y), \textsc{ppl}(x), \textsc{own}(x,y)\right\}]]]$$

b. $\Leftrightarrow$ **at-most-3%**$_u$[**at-least-70%**$_\nu$

$$[\lambda g.\ \left\{g^{u\mapsto x}_{\nu\mapsto y} \middle| \begin{array}{l} y = \iota y[\textsc{land}(y) \wedge \textsc{own}(x,y) \wedge \textsc{max}_{G_{\text{uneven}}}(\langle\%_{\text{ppl}}(x), \%_{\text{land}}(y)\rangle)] \\ x = \iota x[\textsc{ppl}(x) \wedge \textsc{own}(x,y) \wedge \textsc{max}_{G_{\text{uneven}}}(\langle\%_{\text{ppl}}(x), \%_{\text{land}}(y)\rangle)] \end{array}\right\}]]$$

c. $\Leftrightarrow$

$$\lambda g.\ \left\{g^{u\mapsto x}_{\nu\mapsto y} \middle| \begin{array}{l} y = \iota y[\textsc{land}(y) \wedge \textsc{own}(x,y) \wedge \textsc{max}_{G_{\text{uneven}}}(\langle\%_{\text{ppl}}(x), \%_{\text{land}}(y)\rangle)] \\ x = \iota x[\textsc{ppl}(x) \wedge \textsc{own}(x,y) \wedge \textsc{max}_{G_{\text{uneven}}}(\langle\%_{\text{ppl}}(x), \%_{\text{land}}(y)\rangle)] \end{array}\right\}$$

if $\%_{\text{ppl}}(x) \leq 3\% \wedge \%_{\text{land}}(y) \geq 70\%$

(68)

$$\textbf{at-most-3\%}_u \stackrel{\text{def}}{=} \lambda m_{\langle s,st\rangle}.\lambda g_s.\begin{cases} m(g) & \text{if } \%(g(u)) \leq 3\% \\ \varnothing & \text{otherwise} \end{cases}$$

$$\textbf{at-least-70\%}_\nu \stackrel{\text{def}}{=} \lambda m_{\langle s,st\rangle}.\lambda g_s.\begin{cases} m(g) & \text{if } \%(g(\nu)) \geq 70\% \\ \varnothing & \text{otherwise} \end{cases}$$

The derivation in (67) can be paraphrased as follows: people own land (unevenly) (see dref introduction in (67a)), and in the most extreme case (see QUD-based dref selection in (67b)), the owning population is no more than 3% and the land they own is no less than 70% (see the cardinality tests in (67c)).

Similarly, (69) and (70) derive the maximal cumulative readings associated with overall magnitude and a constant rate, respectively.

(69) can be paraphrased as: In the most informative characterization of teaching quality, (so many as) about 30 boring professors gave (so many as) about 50 silly lectures. (70) can be paraphrased as: In the most informative characterization of corn productivity, (so little as) less than 20 acres of farmland yielded (so much as) over 4,000 bushels of corn.

In (70), all relevant drefs yield the same productivity rate, i.e., $\frac{\mu_{\text{CORN}}(y)}{\mu_{\text{LAND}}(x)}$. Thus, with G_{rate} (see (66)), $\mathbf{M}_{\text{QUD-INFO}}$ selects the mereologically maximal drefs.

(69) $[\![$About 50$^{\nu}$ silly lectures were given by about 30^{u} boring professors$]\!]$ (=(51))

$$\Leftrightarrow \textbf{about-30}_{u}[\textbf{about-50}_{\nu}[\mathbf{M}_{\text{QUD-INFO}}[\lambda g.\, \left\{ g^{\nu\mapsto y}_{u\mapsto x} \middle| \text{LECTURE}(y), \text{PROF}(x), \text{GIVE}(x,y) \right\}]]]$$

$$\Leftrightarrow \lambda g.\, \left\{ g^{\nu\mapsto y}_{u\mapsto x} \middle| \begin{array}{l} y = \sigma y[\text{LECTURE}(y) \wedge \text{GIVE}(x,y)] \\ x = \sigma x[\text{PROF}(x) \wedge \text{GIVE}(x,y)] \end{array} \right\}, \text{ if } |x| \approx 30 \wedge |y| \approx 50$$

(70) $[\![$Less than 20^{u} acres (of farmland) yielded over 4,000$^{\nu}$ bushels of corn$]\!]$ (=(52))

$$\Leftrightarrow \textbf{<20 ac}_{u}[\textbf{>4000 bu}_{\nu}[\mathbf{M}_{\text{QUD-INFO}}[\lambda g.\, \left\{ g^{\nu\mapsto y}_{u\mapsto x} \middle| \text{CORN}(y), \text{LAND}(x), \text{YIELD}(x,y) \right\}]]]$$

$$\Leftrightarrow \lambda g.\, \left\{ g^{\nu\mapsto y}_{u\mapsto x} \middle| \begin{array}{l} y = \sigma y[\text{CORN}(y) \wedge \text{YIELD}(x,y)] \\ x = \sigma x[\text{LAND}(x) \wedge \text{YIELD}(x,y)] \end{array} \right\}, \text{ if } \left\{ \begin{array}{l} \mu_{\text{land}}(x) < 20 \text{ ac} \wedge \\ \mu_{\text{corn}}(y) > 4000 \text{ bu} \end{array} \right.$$

Together, the derivations in (67), (69), and (70) capture the intuitive interpretations of these sentences. More generally, all three classes of cumulative readings are derived by the same interpretation mechanism: a QUD-based dref selection parameterized by the composite scale G_{QUD}. Different choices of G_{QUD} correspond to different patterns of interaction among dimensions, reflecting different conversational goals.

4.2 The meaning of a comparative

The above analysis implements the meaning derivation of multi-dimensional measurement sentences in dynamic semantics. Before turning to MHCs, I show that this dynamic framework extends naturally to regular single-dimensional comparatives. Though the semantics of regular comparatives is already well understood (see Beck 2011 for a detailed review), the current dynamic framework derives the same truth conditions while offering greater conceptual unification via independently motivated elements.

In the canonical analysis, as illustrated in (71), a gradable adjective such as tall relates a degree d and an atomic individual x, meaning that along the relevant scale (here height), the measurement of x reaches degree d.

(71) $[\![\text{tall}]\!] \overset{\text{def}}{=} \lambda d.\lambda x.\textsc{height}(x) \geq d$ (i.e., atomic individual x is tall to degree d)

Based on the meaning of *-er* (see (72)) and the assumption of an elided gradable adjective in the *than*-clause, (73) derives the meaning of a comparative sentence. With lambda abstraction, the matrix and *than*-clauses each denote a set of degrees. Then *-er* applies the MAX operator to each set and orders the resulting maximal degrees.

(72) $[\![\text{-er}]\!] \overset{\text{def}}{=} \lambda D_{\textsc{than}\langle dt\rangle}, \lambda D_{\textsc{matrix}\langle dt\rangle}.\textsc{max}(D_{\textsc{matrix}}) > \textsc{max}(D_{\textsc{than}})$
$(\textsc{max} \overset{\text{def}}{=} \lambda D.\iota d[d \in D \wedge \forall d' \in D[d' \leq d]])$

(73) Lucy is taller than $[\lambda d.$ Mary is ~~d-tall~~$]$.
LF: -er $[\lambda d.$Mary is d-tall$][\lambda d.$Lucy is d-tall$]$
$[\![(73)]\!] \Leftrightarrow \textsc{max}(\lambda d.\textsc{height}(\text{Lucy}) \geq d) > \textsc{max}(\lambda d.\textsc{height}(\text{Mary}) \geq d)$
$\Leftrightarrow \textsc{height}(\text{Lucy}) > \textsc{height}(\text{Mary})$.

This canonical analysis works naturally for gradable adjective *tall*, but requires a special treatment for its antonym, *short*. The contrast between (71) and (74) shows that *tall* and *short* order degrees in opposite directions. Therefore, deriving the meaning of a comparative sentence with *short* (see (76)) requires a different lexical entry for the comparative morpheme *-er* (see (75)). The two definitions (see (72) vs. (75)) differ in two respects: (i) whether they compare maximal or minimal degrees, and (ii) whether the comparison uses > or <. Although the correct truth conditions are derived, positing two lexical entries for *-er* is stipulative and theoretically unattractive (see Beck 2011).[18]

[18] The canonical analysis also faces empirical challenges. For example, (i) contains a plural definite DP in its *than*-clause. Rather than comparing Lucy's height with that of the tallest boy(s), as intuition suggests, the canonical analysis compares her height with that of the shortest boy(s), thereby yielding truth conditions that are too weak. To account for such cases, Schwarzschild and Wilkinson (2002), Beck (2010), Zhang (2020), and Zhang and Ling (2021) adopt an interval-based analysis. Comparatives with modified numerals in the *than*-clause (e.g., (ii)) further show that interval semantics is indispensable even under the current dynamic framework (see Zhang 2020). Since interval semantics is orthogonal to the current analysis of MHCs, I adopt degree semantics to avoid unnecessary complications.

(i) Lucy is taller than the boys are ~~tall~~.
LF in the canonical analysis: -er $[\lambda d.$the boys are DIST d-tall$][\lambda d.$Lucy is d-tall$]$
$\rightsquigarrow \textsc{height}(\text{Lucy}) > \textsc{height}(\text{the shortest boy})$ Counterintuitive and truth-conditionally too weak!

(ii) Lucy is taller than exactly three boys. (see Schwarzschild 2008, Zhang 2020)

(74) $[\![\text{short}]\!] \stackrel{\text{def}}{=} \lambda d.\lambda x.\textsc{height}(x) \leq d$

(75) $[\![\text{-er}_{\text{antonym}}]\!] \stackrel{\text{def}}{=} \lambda D_{\textsc{than}\langle dt\rangle}, \lambda D_{\textsc{matrix}\langle dt\rangle}.\textsc{min}(D_{\textsc{matrix}}) < \textsc{min}(D_{\textsc{than}})$
$(\textsc{min} \stackrel{\text{def}}{=} \lambda D.\iota d[d \in D \wedge \forall d' \in D[d' \geq d]])$

(76) Mary is shorter than $[\lambda d.$ Lucy is ~~d-short~~$]$.
LF: $\text{-er}_{\text{antonym}}$ $[\lambda d.$Lucy is d-short$][\lambda d.$Mary is d-short$]$
$[\![(76)]\!] \Leftrightarrow \textsc{min}(\lambda d.\textsc{height}(\text{Mary}) \leq d) < \textsc{min}(\lambda d.\textsc{height}(\text{Lucy}) \leq d)$
$\Leftrightarrow \textsc{height}(\text{Mary}) < \textsc{height}(\text{Lucy})$

By introducing and manipulating drefs, dynamic semantics cleanly separates the roles of indefiniteness and definiteness, which are often lexically encoded together. In the cumulative readings discussed above, modified numerals contribute (i) **indefiniteness** by introducing drefs and (ii) **definiteness** through QUD-based informativeness maximization. The same decomposition extends naturally to comparatives.

As sketched in (77)–(79), the MAX/MIN operators in the lexical entries for *-er* (see (72) and (75)) consist of two components. **First**, a hidden *wh*-item (here *how*) contributes indefiniteness by introducing a degree dref (see Comorovski 1996). **Second**, an answerhood operator contributes definiteness by selecting the most informative dref (see Dayal 1996). Thus, the proposed informativeness maximization operator $\mathbf{M}_{\textsc{qud-info}}$ functions as the answerhood operator. Whether the selected degree dref is maximal or minimal depends on the scale ordering encoded by the relevant gradable adjective.

The remaining difference between (72) and (75) is the ordering relation (> vs. <), reflecting the polarity of the comparative expression: *-er* and *less* denote a positive and a negative difference, respectively. Thus, the dynamic framework replaces the stipulations in (72) and (75) with independently motivated components.

(77) Lucy is taller than $[\lambda d.$ Mary is ~~d-tall~~$]$. An increase in tallness
$\rightsquigarrow$ $\mathbf{Ans}_u[\text{how}^u$ tall Lucy is$]$ $\text{exceed}_{\text{tall}}$ $\mathbf{Ans}_\nu[\text{how}^\nu$ tall Mary is$]$
(cf. (73): $\textsc{max}(\lambda d.\textsc{height}(\text{Lucy}) \geq d) > \textsc{max}(\lambda d.\textsc{height}(\text{Mary}) \geq d)$)

(78) Mary is shorter than $[\lambda d.$ Lucy is ~~d-short~~$]$. An increase in shortness
$\rightsquigarrow$ $\mathbf{Ans}_u[\text{how}^u$ short Mary is$]$ $\text{exceed}_{\text{short}}$ $\mathbf{Ans}_\nu[\text{how}^\nu$ short Lucy is$]$
(cf. (76): $\textsc{min}(\lambda d.\textsc{height}(\text{Mary}) \leq d) < \textsc{min}(\lambda d.\textsc{height}(\text{Lucy}) \leq d)$)

(79) Mary is less tall than $[\lambda d.$ Lucy is ~~d-tall~~$]$. A decrease in tallness
$\rightsquigarrow$ $\mathbf{Ans}_u[\text{how}^u$ tall Mary is$]$ $\text{falls-below}_{\text{tall}}$ $\mathbf{Ans}_\nu[\text{how}^\nu$ tall Lucy is$]$

Thus, under this dynamic view, the meaning of a comparative is derived in three steps. **First**, in each of the matrix and *than*-clauses, a hidden degree dref is introduced and restricted (e.g., in (82), '$\textsc{height}(\text{Lucy}) \geq d_1$' means that Lucy's height is tall to an indefinite degree d_1). **Second**, based on the relevant informativeness measure G_{QUD} (see (80)), $\mathbf{M}_{\text{QUD-INFO}}$ selects the most informative degree drefs (see (63)). **Finally**, comparative morphemes, *-er* and *less* (see (81)), perform post-suppositional comparison tests on the selected degree drefs, indicating a (non-specific or specified) increase (positive difference) or decrease (negative difference) along the relevant scale.

(80) a. $G_{\uparrow}(d) \overset{\text{def}}{=} d$ (for positive adjectives, e.g., *tall*, *big*, *heavy*) $\qquad \uparrow d \Rightarrow \uparrow G_{\uparrow(d)}$

b. $G_{\downarrow}(d) \overset{\text{def}}{=} -d$ (for negative adjectives, e.g., short, small, light) $\qquad \downarrow d \Rightarrow \uparrow G_{\downarrow}(d)$

(81) a. $[\![\text{-er}]\!] = \lambda d_{\text{DIFF}}.\lambda G.\lambda d_1.\lambda d_2.\ \underline{d_{\text{DIFF}} > 0}\ G(d_1) - G(d_2) = d_{\text{DIFF}}$ $\qquad \rightsquigarrow$ **increase**

b. $[\![\text{less}]\!] = \lambda d_{\text{DIFF}}.\lambda G.\lambda d_1.\lambda d_2.\ \underline{d_{\text{DIFF}} < 0}\ G(d_1) - G(d_2) = d_{\text{DIFF}}$ $\qquad \rightsquigarrow$ **decrease**

(82) ⟦Lucy is (about one inch) taller^{u_1} than Mary is ~~tall~~^{u_2}⟧

$$\Leftrightarrow$$

$$\underbrace{(\approx\mathbf{1''})\text{-}\mathbf{er}_{u_1,u_2}}_{\text{3. comparison test}}[\ \underbrace{\mathbf{M}_{\text{QUD-INFO}}}_{\text{2. maximization}}\ \underbrace{[\lambda g.\ \left\{ \overset{u_2\mapsto d_2}{g^{u_1\mapsto d_1}} \middle| \ \textsc{height}(\text{Lucy}) \geq d_1, \textsc{height}(\text{Mary}) \geq d_2 \right\}}_{\text{1. dref introduction}}]]$$

$$\Leftrightarrow \lambda g.\ \left\{ \overset{u_2\mapsto d_2}{g^{u_1\mapsto d_1}} \middle| \begin{array}{l} d_2 = \textsc{height}(\text{Mary}) \\ d_1 = \textsc{height}(\text{Lucy}) \end{array} \right\}, \text{if } G_{\text{tall}}(g(u_1)) - G_{\text{tall}}(g(u_2)) \approx 1'' \text{ (or} > 0)$$

(83) ⟦Mary is (about one inch) shorter^{u_1} than Lucy is ~~short~~^{u_2}⟧

$$\Leftrightarrow (\approx\mathbf{1''})\text{-}\mathbf{er}_{u_1,u_2}[\mathbf{M}_{\text{QUD-INFO}}[\lambda g.\ \left\{ \overset{u_2\mapsto d_2}{g^{u_1\mapsto d_1}} \middle| \ \textsc{height}(\text{Mary}) \leq d_1, \textsc{height}(\text{Lucy}) \leq d_2 \right\}]]$$

$$\Leftrightarrow \lambda g.\ \left\{ \overset{u_2\mapsto d_2}{g^{u_1\mapsto d_1}} \middle| \begin{array}{l} d_2 = \textsc{height}(\text{Lucy}) \\ d_1 = \textsc{height}(\text{Mary}) \end{array} \right\}, \text{if } G_{\text{short}}(g(u_1)) - G_{\text{short}}(g(u_2)) \approx 1'' \text{ (or} > 0)$$

(84) ⟦Mary is (about one inch) less tall^{u_1} than Lucy is ~~tall~~^{u_2}⟧

$$\Leftrightarrow (\approx\mathbf{1''})\text{-}\mathbf{less}_{u_1,u_2}[\mathbf{M}_{\text{QUD-INFO}}[\lambda g.\ \left\{ \overset{u_2\mapsto d_2}{g^{u_1\mapsto d_1}} \middle| \ \textsc{height}(\text{Mary}) \geq d_1, \textsc{height}(\text{Lucy}) \geq d_2 \right\}]]$$

$$\Leftrightarrow \lambda g.\ \left\{ \overset{u_2\mapsto d_2}{g^{u_1\mapsto d_1}} \middle| \begin{array}{l} d_2 = \textsc{height}(\text{Lucy}) \\ d_1 = \textsc{height}(\text{Mary}) \end{array} \right\}, \text{if } G_{\text{tall}}(g(u_1)) - G_{\text{tall}}(g(u_2)) \approx -1'' \text{ (or} < 0)$$

Consequently, (82) is true iff there is a height increase from Mary's to Lucy's; (83) is true iff there is a shortness increase from Lucy's to Mary's; and (84) is true iff there is a height decrease from Lucy's to Mary's. These derivations capture the intuitive readings in a principled way.

4.3 The meaning of an MHC

Based on the derivations of cumulative-reading sentences (§4.1) and comparatives (§4.2), the analysis of MHCs follows straightforwardly. (85)–(88) illustrate the three classes of MHCs, all derived by the same mechanism, and the only distinction is the parameterized choice of G_{QUD}.

With G_{uneven} (see (64)), the comparisons in (85) are performed on the drefs that most informatively characterize the unevenness of wealth distribution. (85) is true iff the selected drefs exhibit a decrease in population share and an increase in wealth share. Together, these changes yield a higher G_{uneven} for the matrix than the *than*-clause, expressing an increase in unevenness from elsewhere to here (see Fig. 6).[19]

(85) ⟦Fewer^{u_1,m_1} people own more$^{\nu_1,n_1}$ wealth here
than [~~many^{u_2,m_2} people own much$^{\nu_2,n_2}$ wealth~~ elsewhere]⟧ $\rightsquigarrow G_{\text{uneven}}$ (see (64))

a. $\Leftrightarrow$

$$\textbf{-er}_{m_1,m_2}[\textbf{-er}_{n_1,n_2}[\textbf{M}_{\text{QUD-INFO}}[\lambda g.\left\{ g^{\substack{m_1\mapsto d_{\text{ppl1}}\\ n_1\mapsto d_{\text{wth1}}\\ \nu_1\mapsto y_1\\ u_1\mapsto x_1\\ m_2\mapsto d_{\text{ppl2}}\\ n_2\mapsto d_{\text{wth2}}\\ \nu_2\mapsto y_2\\ u_2\mapsto x_2}} \;\middle|\; \begin{array}{l} \textsc{ppl-here}(x_1), \textsc{wealth-here}(y_1),\\ \textsc{own}(x_1, y_1)\\ \textsc{ppl-else}(x_2), \textsc{wealth-else}(y_2),\\ \textsc{own}(x_2, y_2)\\ \%_{\text{ppl}}(x_1) \geq d_{\text{ppl1}}, \%_{\text{wealth}}(y_1) \geq d_{\text{wth1}}\\ \%_{\text{ppl}}(x_2) \geq d_{\text{ppl2}}, \%_{\text{wealth}}(y_2) \geq d_{\text{wth2}} \end{array} \right\}]]]$$

b.
$$\Leftrightarrow \lambda g.\left\{ g^{\substack{m_1\mapsto d_{\text{PPL1}}\\ n_1\mapsto d_{\text{WTH1}}\\ \nu_1\mapsto y_1\\ u_1\mapsto x_1\\ m_2\mapsto d_{\text{PPL2}}\\ n_2\mapsto d_{\text{WTH2}}\\ \nu_2\mapsto y_2\\ u_2\mapsto x_2}} \;\middle|\; \begin{array}{l} x_1 = \iota x_1[\textsc{ppl-here}(x_1) \wedge \textsc{own}(x_1, y_1) \wedge\\ \textsc{max}_{G_{\text{uneven}}}(\langle \%_{\text{ppl}}(x_1), \%_{\text{wealth}}(y_1)\rangle)]\\ y_1 = \iota y_1[\textsc{wealth-here}(y_1) \wedge \textsc{own}(x_1, y_1) \wedge\\ \textsc{max}_{G_{\text{uneven}}}(\langle \%_{\text{ppl}}(x_1), \%_{\text{wealth}}(y_1)\rangle)]\\ x_2 = \iota x_2[\textsc{ppl-else}(x_2) \wedge \textsc{own}(x_2, y_2) \wedge\\ \textsc{max}_{G_{\text{uneven}}}(\langle \%_{\text{ppl}}(x_2), \%_{\text{wealth}}(y_2)\rangle)]\\ y_2 = \iota y_2[\textsc{wealth-else}(y_2) \wedge \textsc{own}(x_2, y_2) \wedge\\ \textsc{max}_{G_{\text{uneven}}}(\langle \%_{\text{ppl}}(x_2), \%_{\text{wealth}}(y_2)\rangle)]\\ d_{\text{ppl1}} = \%_{\text{ppl}}(x_1), d_{\text{wth1}} = \%_{\text{wealth}}(y_1)\\ d_{\text{ppl2}} = \%_{\text{ppl}}(x_2), d_{\text{wth2}} = \%_{\text{wealth}}(y_2) \end{array} \right\},$$

$$\text{if} \begin{cases} G_{\text{many}}(g(m_1)) - G_{\text{many}}(g(m_2)) < 0 \wedge \\ G_{\text{much}}(g(n_1)) - G_{\text{much}}(g(n_2)) > 0 \end{cases} \quad (\text{i.e., } d_{\text{ppl1}} < d_{\text{ppl2}} \wedge d_{\text{wth1}} > d_{\text{wth2}})$$

[19] In (85), analyzing *fewer* as '*less* + *many*' or '*few* + *-er*' yields the same derivation (see (78) and (79)).

With G_{mag} (see (65)), the comparisons in (86) are performed on the drefs that most informatively characterize the quality of teaching. (86) is true iff the selected drefs exhibit cardinality increases in both silly lectures and boring professors. Together, these changes yield a higher G_{mag} for the matrix than the *than*-clause, expressing a deterioration in teaching quality from expectation to reality (see Fig. 7).

(86) $[\![$More$^{\nu_1,n_1}$ silly lectures have been given by more^{u_1,m_1} boring professors than [I would have expected ~~many$^{\nu_2,n_2}$ silly lectures have been given by many^{u_2,m_2} boring professors~~]$]\!]$. $\rightsquigarrow G_{\text{mag}}$ (see (65))

$$\Leftrightarrow \lambda g.\left\{ g^{\begin{smallmatrix} m_1\mapsto d_{\text{prof1}} \\ n_1\mapsto d_{\text{lec1}} \\ \nu_1\mapsto y_1 \\ u_1\mapsto x_1 \\ m_2\mapsto d_{\text{prof2}} \\ n_2\mapsto d_{\text{lec2}} \\ \nu_2\mapsto y_2 \\ u_2\mapsto x_2 \end{smallmatrix}} \;\middle|\; \begin{array}{l} x_1 = \sigma x_1[\textsc{prof}(x_1) \wedge \textsc{give}(x_1, y_1)] \\ y_1 = \sigma y_1[\textsc{lecture}(y_1) \wedge \textsc{give}(x_1, y_1)] \\ x_2 = \sigma x_2[\textsc{prof}(x_2) \wedge \textsc{give}(x_2, y_2)] \\ y_2 = \sigma y_2[\textsc{lecture}(y_2) \wedge \textsc{give}(x_2, y_2)] \\ d_{\text{prof1}} = \textsc{card}(x_1), d_{\text{lec1}} = \textsc{card}(y_1) \\ d_{\text{prof2}} = \textsc{card}(x_2), d_{\text{lec2}} = \textsc{card}(y_2) \end{array} \right\},$$

$$\text{if} \begin{cases} G_{\text{many}}(g(m_1)) - G_{\text{many}}(g(m_2)) > 0 \wedge \\ G_{\text{many}}(g(n_1)) - G_{\text{many}}(g(n_2)) > 0 \end{cases} \quad (\text{i.e., } d_{\text{prof1}} > d_{\text{prof2}} \wedge d_{\text{lec1}} > d_{\text{lec2}})$$

With G_{rate} (see (66)), the comparisons in (87) are performed on the drefs that most informatively characterize corn productivity. (87) is true iff the selected drefs exhibit a decrease in farmland and an increase in corn yield. Together, these changes yield a higher G_{rate} for the matrix than the *than*-clause, expressing an increase in corn productivity from before to now (see Fig. 8).

(87) $[\![$Less^{u_1,m_1} land produces more$^{\nu_1,n_1}$ corn than [~~much^{u_2,m_2} land produced much$^{\nu_2,n_2}$ corn~~ ever before]$]\!]$. $\rightsquigarrow G_{\text{rate}}$ (see (66))

$$\Leftrightarrow \lambda g.\left\{ g^{\begin{smallmatrix} m_1\mapsto d_{\text{land1}} \\ n_1\mapsto d_{\text{corn1}} \\ \nu_1\mapsto y_1 \\ u_1\mapsto x_1 \\ m_2\mapsto d_{\text{land2}} \\ n_2\mapsto d_{\text{corn2}} \\ \nu_2\mapsto y_2 \\ u_2\mapsto x_2 \end{smallmatrix}} \;\middle|\; \begin{array}{l} x_1 = \sigma x_1[\textsc{land}(x_1) \wedge \textsc{produce}(x_1, y_1)] \\ y_1 = \sigma y_1[\textsc{corn}(y_1) \wedge \textsc{produce}(x_1, y_1)] \\ x_2 = \sigma x_2[\textsc{land}(x_2) \wedge \textsc{produce}(x_2, y_2)] \\ y_2 = \sigma y_2[\textsc{corn}(y_2) \wedge \textsc{produce}(x_2, y_2)] \\ d_{\text{land1}} = \mu_{\text{land}}(x_1), d_{\text{corn1}} = \mu_{\text{corn}}(y_1) \\ d_{\text{land2}} = \mu_{\text{land}}(x_2), d_{\text{corn2}} = \mu_{\text{corn}}(y_2) \end{array} \right\},$$

$$\text{if} \begin{cases} G_{\text{much}}(g(m_1)) - G_{\text{Much}}(g(m_2)) < 0 \wedge \\ G_{\text{much}}(g(n_1)) - G_{\text{much}}(g(n_2)) > 0 \end{cases} \quad (\text{i.e., } d_{\text{land1}} < d_{\text{land2}} \wedge d_{\text{corn1}} > d_{\text{corn2}})$$

Finally, again with G_{rate} (see (66)), the comparisons in (88) are performed on the drefs that most informatively characterize value for money. (88) is true iff the selected drefs exhibit a small increase in quality but a large increase in price. Together, these changes yield a lower G_{rate} for the matrix than the *than*-clause, expressing a decrease in value for money from last year to this year (see Fig. 9).

(88) $[\![$Slightly better^{m_1} products cost a lot more^{n_1} than [~~good^{m_2} products cost much^{n_2}~~ last year]$]\!]$. $\rightsquigarrow G_{\text{rate}}$ (see (66))

$$\Leftrightarrow \lambda g \,.\left\{ g^{\substack{m_1 \mapsto d_{\text{q1}} \\ n_1 \mapsto d_{\text{price1}} \\ m_2 \mapsto d_{\text{q2}} \\ n_2 \mapsto d_{\text{price2}}}} \;\middle|\; \begin{array}{l} d_{\text{q1}} = \mu_{\text{quality}}(\text{products-of-this-year}), \\ d_{\text{price1}} = \mu_{\text{price}}(\text{products-of-this-year}), \\ d_{\text{q2}} = \mu_{\text{quality}}(\text{products-of-last-year}), \\ d_{\text{price2}} = \mu_{\text{price}}(\text{products-of-last-year}) \end{array} \right\},$$

$$\text{if} \begin{cases} G_{\text{good}}(g(m_1)) - G_{\text{good}}(g(m_2)) = d_{\text{small-difference-of-quality}} \wedge \\ G_{\text{much}}(g(n_1)) - G_{\text{much}}(g(n_2)) = d_{\text{large-difference-of-price}} \end{cases}$$

The example in (88) is particularly interesting because the underlying meaning, a decrease in value for money, is conveyed not by opposite directions of change, but by different magnitudes of change in the same direction. Both quality and price increase, but the increase in price outweighs the increase in quality. This observation motivates characterizing comparisons in terms of positive and negative differences (see (81)), rather than ordering relations $>$ and $<$ (see (72) and (75)), because differences encode both the direction and the magnitude of change (see also §5.3).

Moreover, the same pattern of co-directional changes with different magnitudes should also arise in MHCs addressing changes in unevenness. The current analysis therefore predicts that (89) is true in the context below.

(89) With advances in technology, a slightly larger share of the population owns a much larger share of the total wealth than before.
(Context: The group of people holding a large share of the wealth has grown only slightly, from 2% to 3% of the population, while its share of the total wealth has increased substantially, from 60% to 85%.)

Overall, the derivations in (85)–(89) capture the intuitive readings discussed in §3.3 and demonstrate that MHCs arise naturally from the combination of cumulative readings and comparatives.

5 Discussion

The current analysis of MHCs offers further insights into the role of hidden degree QUDs in natural language (§5.1), the pragmatic source of maximal informativeness and its role in semantic composition (§5.2), and the nature of multi-dimensional comparison (§5.3).

5.1 Hidden degree QUDs

Following Krifka (1999)'s view of cumulative readings (see also Zhang 2023), the proposed analysis of MHCs shows that the interaction between numerals or comparatives reflects interlocutors' underlying conversational goal. Specifically, this conversational goal can be characterized as a degree QUD. This perspective connects MHCs to other phenomena whose interpretation likewise involves hidden degree QUDs: e.g., focus sensitive particle *even* (e.g., (90); Greenberg 2018, Zhang 2022) and over-/under-statements (i.e., hyperbole and meiosis, see (91); Nouwen 2024).

(90) (Context: Imagine Pooh and friends coming upon a bush of thistles. Eeyore, who is known to favor thistles, takes a bite but spits it out. (Szabolcsi 2017))
Those thistles must be really prickly! <u>Even</u> [Eeyore]$_F$ spit them out!

(91) a. About a party: '<u>A hundred people</u> came.' **Overstatement / hyperbole**
b. About a party: '<u>Nobody</u> came.' **Understatement / meiosis**

The primary communicative purpose of (90) and (91) is not to identify an individual or report a number literally. Instead, they express the prickliness of the thistles or the (un)success of the party.

One piece of evidence comes from patterns of question-answer congruence (Zhang 2022). As illustrated in (92) and (93), an *even*-sentence cannot felicitously answer a *wh*-question targeting the focus associate of *even*, here *Eeyore*. The incongruence in (93) also challenges the canonical view that *even* involves a likelihood scale and marks the individual corresponding to low likelihood (Karttunen and Peters 1979).

(92) Q: Who spit the thistles out? (Under the scenario in (90))
A: #Even [Eeyore]$_F$ spit them out. The presence of *even* is infelicitous

(93) Q: Who was unlikely to spit the thistles out?
A: #Even [Eeyore]$_F$ spit them out. (In (90), only Eeyore spit the thistles out)

The question-answer congruence in (94) suggests that an *even*-sentence addresses a hidden degree QUD. Motivated by these empirical observations, Zhang (2022) proposes a degree-QUD-based analysis of *even* (see (95)). On this view, the prejacent of *even* (e.g., *Eeyore spit them out* in (90)) is maximally informative among the alternatives (e.g., *Pooh spit them out*) in resolving the hidden degree QUD (here *how prickly the thistles are*).

(94) Q: How prickly are those thistles? — A degree question
A: Oh, even [Eeyore]$_F$ spit them out. — Felicitous answer in the scenario in (90)
Paraphrase: those thistles are prickly to such a degree that Eeyore spit them out.

(95) The felicity of *even*(p) requires: (Zhang 2022; see also Greenberg 2018)
a. there is a contextually salient degree QUD, and
b. the prejacent p is maximally informative in resolving the degree QUD

Similarly, hyperbole and meiosis communicate an extreme degree on a contextually relevant scale that captures the interlocutors' concern (see (96); see Nouwen 2024).

(96) a. (91a) $\rightsquigarrow$ The party was so successful that it was as if a hundred people came.
b. (91b) $\rightsquigarrow$ The party was such a failure that it was as if nobody came.

Hidden degree QUDs therefore underlie a range of linguistic phenomena, including MHCs, cumulative readings, *even*, and over-/under-statements.

5.2 Maximal informativeness

A further issue is how degree QUDs and maximal informativeness contribute to sentence interpretation. The canonical view assumes that the uttered sentence is the maximally informative true member of a relevant set of alternatives (see (97) and (98)).

(97) [At most 3%]$_F$ of the population own [at least 70%]$_F$ of the land.

$$\left\{\begin{matrix} \dots \\ \leq 2\% \\ \underline{\leq 3\%} \\ \leq 4\% \\ \dots \end{matrix}\right\} \text{ of the population own } \left\{\begin{matrix} \dots \\ \geq 60\% \\ \underline{\geq 70\%} \\ \geq 80\% \\ \dots \end{matrix}\right\} \text{ of the land.}$$

(98) Lucy weighs [exactly 55]$_F$ kg now.

$$\Big\{\dots, \textsc{Weight}(\text{Lucy}) = 50 \text{ kg}, \underline{\textsc{Weight}(\text{Lucy}) = 55 \text{ kg}}, \textsc{Weight}(\text{Lucy}) = 60 \text{ kg}, \dots\Big\}$$

However, determining the maximally informative member from the above alternative sets (see (97) and (98)) poses two technical challenges.

First, cumulative readings such as (97) require the two modified numerals to be interpreted jointly: informativeness decreases along one dimension while increasing along the other. Thus, informativeness cannot be determined by considering either numeral in isolation, and the joint interpretation requires an additional mechanism.

Second, non-monotonic modified numerals, as in (98), give rise to alternatives that do not stand in entailment relations, making it unclear how a maximally informative alternative should be identified.

The post-suppositional analysis adopted here (Brasoveanu 2013; see also e.g., Brasoveanu and Szabolcsi 2012 for discussion) solves both problems.

First, maximal informativeness is determined by the underlying degree QUD (Beck and Rullmann 1999), rather than by entailment among alternatives. In this sense, pragmatic reasoning does not come into play only after bottom-up semantic composition is complete. Instead, the degree QUD guides semantic composition in a top-down way, by determining the informativeness ordering, as illustrated by (99).

(99)
a. Interpreting (98) in a growth context: maximal informativeness is achieved by $\textsc{max}[\lambda d.\textsc{height}(\text{Lucy}) \geq d]$, i.e., $\uparrow d \Rightarrow \uparrow$ informativeness
b. Interpreting (98) in a weight-loss context: maximal informativeness is achieved by $\textsc{min}[\lambda d.\textsc{height}(\text{Lucy}) \leq d]$, i.e., $\downarrow d \Rightarrow \uparrow$ informativeness

Second, QUD-based maximal informativeness does not directly select a sentence from the alternative set (as shown in (97) and (98)). Instead, it serves to define the analytically true (i.e., tautological) answer to the QUD: e.g., 'the population and land ownership corresponding to maximal unevenness' in addressing *how uneven wealth distribution is*, 'Lucy's weight' in addressing *how much Lucy weighs*.[20] Additional information about this analytic answer is then provided (e.g., *exactly 55* in (98)).

This understanding of maximal informativeness not only resolves the above technical challenges, but also explains the difference between measure sentences and *even*-sentences, as illustrated by (100).

[20] See also the parallelism between questions and concealed questions (e.g., Nathan 2006, Barker 2016). A question, its corresponding concealed question (see (i)), and the corresponding analytic answer all encode the same information. See also Dayal (1996) on maximal informativeness in answerhood.

(i)
a. I know how much Lucy weighs.
b. I know the weight of Lucy. — A concealed question

(100) a. Bill has [three]$_F$ kids. QUD: how many kids does Bill have?
$\rightsquigarrow$ [$\underbrace{\text{The cardinality of the totality of kids Bill has}}_{\text{the analytic answer to the QUD}}$] is (at least) 3.

b. Bill even has [three]$_F$ kids. sample QUD: how heavy is Bill's burden?
$\rightsquigarrow$ [$\underbrace{\text{Bill's burden is heavy to such a degree}}_{\text{the analytic answer to the QUD}}$] that he has three kids.

Under the canonical Gricean and focus-semantic view, (100a) and (100b) contain the same focused item and therefore invoke the same alternative set. If sentence interpretation is determined solely by selecting the maximally informative true alternative, the two sentences are predicted to receive the same interpretation. Contrary to this prediction, however, they are intuitively interpreted quite differently.

Under the current approach, (100a) and (100b) not only address different degree QUDs (as reflected by what questions they felicitously answer), but also differ in what constitutes (i) the analytic answer and (ii) the additional information.

In (100a), the QUD targets the cardinality of the mereologically largest dref (i.e., the totality of kids Bill has), and *three* specifies this cardinality.

In (100b), by contrast, the QUD targets the degree of heaviness reached by Bill's burden, and the entire prejacent of *even* characterizes this degree dref. Moreover, under the requirement imposed by *even* (see (95), §5.1), this prejacent identifies the maximal (i.e., highest) degree of heaviness in the context, so that alternatives like *Bill has four kids* do not characterize a higher degree of heaviness.

This analysis therefore captures the distinct interpretations of (100a) and (100b) while preserving a unified notion of maximal informativeness. More generally, this view of maximal informativeness is both technically more convenient and explanatorily more powerful than the canonical view.[21]

[21] The current view also predicts and explains another difference between (100a) and (100b): while (100b) is norm-sensitive (yielding the inference that Bill's burden is heavy), (100a) is not. This follows naturally from the prejacent's role in characterizing the maximal degree of heaviness on the relevant scale: this maximal degree must exceed the expected norm (see Krifka 2000).

Under the current view, MHCs and cumulative-reading sentences are expected to pattern with regular comparatives and measure sentences in lacking norm-sensitivity (see e.g., Rett 2015 on the norm-sensitivity of gradable expressions). However, some MHCs, especially those addressing unevenness (e.g., *fewer people own more wealth in this country than anywhere else*), appear to be norm-sensitive, suggesting that the wealth distribution is unusually uneven in this country (see also Haslinger and Schmitt 2020 on the 'surprising effect' of the non-maximal cumulative reading). This is reminiscent of observations on antonyms: *how short Lucy is* is norm-sensitive, whereas *how tall Lucy is* is not. This suggests that norm-sensitive inferences depend, at least in part, on the underlying degree QUD. I leave a full investigation to future work.

5.3 Comparison and numerical differentials

The current study of MHCs favors the view that comparisons are better understood in terms of increases and decreases (i.e., positive and negative differences), which encode both the polarity and magnitude of change (see (81) as well as §4.2 and 4.3).[22] Specifically, in MHCs and cumulative-reading sentences, polarity (e.g., (85)–(87)) and relative magnitude (e.g., (88)) of changes across dimensions provide clues to the underlying composite scale and its informativeness ordering.

In regular, single-dimensional comparatives, numerical differentials are often included, measuring the size of increases or decreases:

(101) a. Mary is 5′8″ tall. Lucy is <u>3 inches taller</u> than Mary is.
b. Lucy is 5′11″ tall. Mary is <u>3 inches less tall</u> than Lucy is.

By contrast, MHCs generally disfavor numerical differentials or explicit quantity specifications,[23] as illustrated by the following contrasts:

(102) Context: Last year, 20 acres yielded 4,000 bushels of corn. This year, 18 acres yielded 4,500 bushels.
a. ??Two fewer acres yielded 500 more bushels of corn than last year.
b. Less land yielded more corn than last year.

(103) Context: I had expected 20 silly lectures to be given by 10 boring professors. Instead, 28 silly lectures were given by 15 boring professors.

[22] See also Greenberg (2010), Zhang and Ling (2021), and Zhang and Zhang-Yukun (2025), which draw parallels between comparison and additivity and analyze comparative morpheme *-er*/*more* alongside additive particles like *other*. The anaphoricity involved in comparison (see §2.3.3; see Li 2023) also aligns with the anaphoricity involved in additivity (see e.g., Beaver and Clark 2009). Examples in (i) are parallel to (101).

(i) Mary bought *Moby Dick* and *Faust*. Lucy bought <u>two {more / other}</u> books.

[23] Oda (2008) provides the following Japanese example, showing that MHCs can include explicit quantity specifications for the comparison standard. Moreover, with the use of *sorezore* ('each'), this example has a distributive reading, unlike the MHCs discussed in this paper. Whether and how these differences reflect a broader cross-linguistic contrast remains an open question that I leave for future research.

(i) San-biki-no neko-ga **sorezore** yon-hiki-no hatukanezumi-o tabeta <u>yorimo</u> (<u>motto</u>) <u>takusanno</u>
3-CL-GEN cat-NOM each 4-CL-GEN mouse-ACC ate THAN (more) many
inu-ga **sorezore** (<u>motto</u>) <u>takusanno</u> dobunezumi-o tabeta.
dog-NOM each (more) many rat-ACC ate
Lit. 'More dogs ate more rats each than three cats ate four mice each.'
⇝ There are 3 cats and each of them ate 4 mice. There are more than 3 dogs and each of them ate more than 4 rats. (Oda 2008: (62) and (63))

a. ??Eight more lectures were given by five more professors than I had expected.

b. More lectures were given by more professors than I had expected.

One possible explanation is that interactions among numerical differentials across dimensions are less salient than interactions in polarity or relative magnitude (e.g., *slightly better* vs. *much more* in (88)) for identifying the composite scale and its informativeness ordering. Consequently, multidimensional numerical differentials rarely serve to reveal a salient conversational goal.

By contrast, conjunctions of single-dimensional comparatives readily allow numerical differentials (see (104)), since each comparative independently expresses a change along its own dimension. Their interpretation therefore does not require constructing a single composite scale or resolving a single hidden degree QUD.

(104) Context: Orchid is composed of 50% red and 50% blue, whereas magenta is composed of 70% red and 30% blue. (see also Coppock 2022)
Magenta has 20% more red and 20% less blue than orchid. **Conjunction**

Therefore, the degraded acceptability of numerical differentials in MHCs provides further evidence that MHCs should not be reduced to two independent comparisons. Rather, they express a unified comparison organized around a single hidden degree QUD and its associated composite scale.

6 Conclusion

This paper draws on insights from comparatives and cumulative-reading sentences to investigate the semantics and pragmatics of MHCs. MHCs express changes along multiple dimensions, whose interplay defines a composite scale and addresses a single underlying degree QUD. This perspective captures both their shared properties with regular comparatives and cumulative readings and their distinctive properties, including sensitivity to contextual goals and the role of maximal informativeness behind what is under comparison. More broadly, the current study highlights how pragmatics can guide semantic composition and how hidden conversational goals can be revealed through explicit linguistic cues.

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